

U1 snRNP regulates chromatin retention of noncoding RNAs

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Long noncoding RNAs (lncRNAs) and promoter- or enhancer-associated unstable transcripts locate preferentially to chromatin, where some regulate chromatin structure, transcription and RNA processing^{1–13}. Although several RNA sequences responsible for nuclear localization have been identified—such as repeats in the lncRNA *Xist* and Alu-like elements in long RNAs^{14–16}—how lncRNAs as a class are enriched at chromatin remains unknown. Here we describe a random, mutagenesis-coupled, high-throughput method that we name ‘RNA elements for subcellular localization by sequencing’ (mutREL-seq). Using this method, we discovered an RNA motif that recognizes the U1 small nuclear ribonucleoprotein (snRNP) and is essential for the localization of reporter RNAs to chromatin. Across the genome, chromatin-bound lncRNAs are enriched with 5′ splice sites and depleted of 3′ splice sites, and exhibit high levels of U1 snRNA binding compared with cytoplasm-localized messenger RNAs. Acute depletion of U1 snRNA or of the U1 snRNP protein component SNRNP70 markedly reduces the chromatin association of hundreds of lncRNAs and unstable transcripts, without altering the overall transcription rate in cells. In addition, rapid degradation of SNRNP70 reduces the localization of both nascent and polyadenylated lncRNA transcripts to chromatin, and disrupts the nuclear and genome-wide localization of the lncRNA *Malat1*. Moreover, U1 snRNP interacts with transcriptionally engaged RNA polymerase II. These results show that U1 snRNP acts widely to tether and mobilize lncRNAs to chromatin in a transcription-dependent manner. Our findings have uncovered a previously unknown role of U1 snRNP beyond the processing of precursor mRNA, and provide molecular insight into how lncRNAs are recruited to regulatory sites to carry out chromatin-associated functions.

To identify *cis* elements that contribute to the localization of RNA to chromatin, we developed and performed REL-seq screens with saturated fragment coverage of nine representative lncRNA and mRNA transcripts in mouse and human cells (Extended Data Fig. 1a–f and Supplementary Note 1). The strategy involves expressing a randomly fragmented RNA sequence alone or fusing it with a minigene encoding green fluorescent protein (GFP), and then analysing its subcellular location through sequencing. We detected a total of 26 chromatin-enriched RNA fragments (enChrs; $P < 0.05$), mainly 50–500 nucleotides in length, in the sense orientation of the host chromatin-associated RNAs but not in cytoplasm-located mRNAs (Extended Data Fig. 2 and Supplementary Tables 1, 2). To uncover key residues that contribute to RNA localization, we chose a 162-nucleotide *NXF1*-enChr, identified from an isoform of *NXF1* in which the introns are retained in the final mRNA, for random mutagenesis followed by REL-seq (mutREL-seq; Extended Data Figs. 1b, 2g). Out of 23 mutations with impaired chromatin binding, 19 are located in a loop region (positions 39–45) containing 7 nucleotides, which, together with 2 upstream nucleotides, comprise a U1-recognition site that base pairs perfectly with the 9-nucleotide 5′ targeting sequence of U1 snRNA (Fig. 1a, b, Extended

Data Fig. 1g–j and Supplementary Note 2). The majority of enChrs overlap with predicted U1-recognition sites and/or exhibit strong binding signals of U1 snRNA, as shown by RNA affinity purification followed by RNA sequencing (RAP-RNA-seq)¹⁷, except for repeat-associated enChrs in *Xist* (Extended Data Figs. 1k, 2, Supplementary Table 2 and Supplementary Discussion 1).

U1 snRNP defines the 5′ splice sites of pre-mRNAs and initiates spliceosome assembly at introns—a process that involves sequential recruitments of U1, U2 and then U4/U6–U5 tri-snRNPs¹⁸. It has been reported that the 5′ splice site regulates the nuclear retention of a handful of mRNAs^{19,20}. To test a role of U1 recognition in RNA–chromatin retention, we constructed GFP reporters that harbour U1 motifs but lack the 3′ splice site, in vectors containing either a polyadenylation signal (PAS) or a *Malat1* 3′-end sequence (Extended Data Fig. 3a, b). The *Malat1* 3′-end sequence stabilizes GFP RNA through a triple-helix RNA structure^{21,22}, thus bypassing the inhibitory effect of U1 snRNP on polyadenylation and RNA stability²³. The wild-type but not mutant U1-targeting sequences promoted the chromatin association of GFP in both the PAS and the 3′ *Malat1* reporters (Extended Data Fig. 3c–g and Supplementary Note 3). In addition, *NXF1*-enChr RNA captured core

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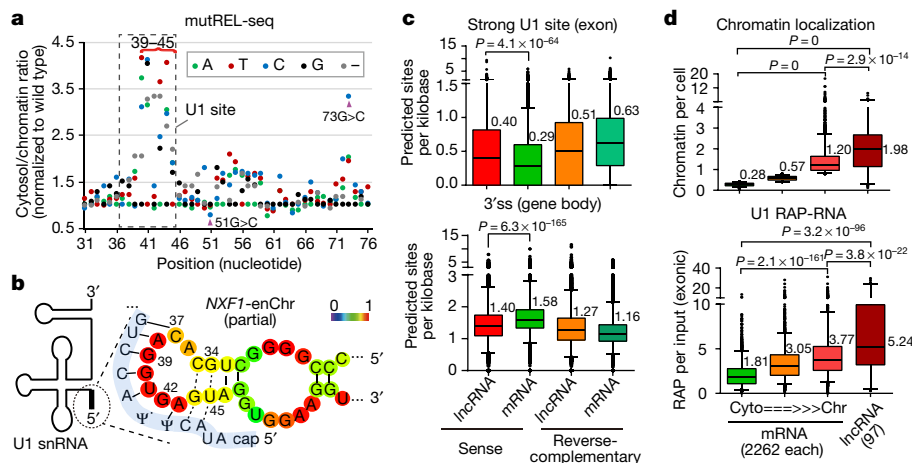


Fig. 1 | MutREL-seq identifies a U1-recognition site that contributes to RNA-chromatin tethering. **a**, MutREL-seq identifies a seven-nucleotide region (dashed box) that contributes to chromatin retention of *NXF1*-enChr. The y-axis shows the fold change of cytoplasmic versus chromatin signals, normalized to that in wild-type cells. Grey dots indicate deletions. **b**, Predicted secondary structure of *NXF1*-enChr and its interaction with U1 snRNA. Ψ:G wobble base-pairing is indicated by dots. The scale bar shows the probability of base-pairing (or lack of pairing). **c**, Densities of predicted strong U1-recognition sites in exons (top) and 3' splice sites (3'ss) in the gene body

(bottom) of lncRNAs (U1 site: $n=3,385$; 3'ss: $n=7,661$) and mRNAs (U1 site: $n=47,298$; 3'ss: $n=78,227$) in mice. Genes with transcript length (for U1 sites) or genomic length (for 3'ss) of more than 1 kilobase (kb) were analysed. **d**, Enrichment in chromatin localization (top) and U1 RAP-RNA signals (bottom) in mRNAs ($n=2,262$ genes in each group) and lncRNAs ($n=97$; fragments per kilobase per million mapped reads (FPKM) of more than 1; no overlap with mRNA) that are ranked from low to high levels of chromatin association (left to right). For **c** and **d**, box plots show 5th, 25th, 50th, 75th and 95th percentiles, with median values labelled. P -values, two-sided t -tests.

components of U1 snRNP, whose knockdown impaired the chromatin binding of reporter RNA (Extended Data Fig. 3h–m, Supplementary Table 3 and Supplementary Note 4). To test a role of the 3' splice site in RNA localization, we inserted an *ACTB* intron into a *GFP* reporter. Mutation of the 3' splice site dramatically increased chromatin-bound *GFP* RNA, in a manner dependent on an intact 5' U1 site (Extended Data Fig. 3n). These results demonstrate that U1 snRNP promotes the chromatin association of a reporter RNA that harbours a 5' U1-recognition site but lacks the 3' splice sequence.

Although U1 motifs tend to be depleted in gene exons, lncRNA transcripts exhibit substantially higher densities of the U1-recognition motif in exons, yet lower densities of 3' splice sites in the gene body, compared with mRNA (Fig. 1c and Extended Data Fig. 4a–d). In addition, the densities of predicted U1-recognition sites and levels of U1 snRNA binding increase gradually from the most cytoplasm-enriched mRNA to chromatin-enriched mRNA and then to lncRNAs, whereas their expression decreases with increased chromatin association (Fig. 1d and Extended Data Fig. 4e, f).

The global enrichment of U1 snRNA recognition sites and binding on lncRNAs led us to explore their role in the localization of lncRNAs to chromatin. We used antisense morpholino oligonucleotides (AMOs) to block the 5'-end-recognition sequence of U1 snRNA. Given the broad involvement of U1 snRNP in cellular functions, we performed a short-term, 2-h treatment with U1 AMOs in mouse embryonic stem cells (mESCs). This attenuated splicing—unlike prolonged treatment—did not elicit apoptosis (Extended Data Fig. 5a–d and Methods). We then performed strand-specific sequencing of total RNAs isolated from chromatin, nucleoplasm and cytoplasm fractions, and calculated relative enrichments in each compartment. Among 1,282 lncRNAs with detectable expression in mESCs, only 4 were upregulated on chromatin, whereas 337 (26.3%; $P < 0.05$) showed decreased chromatin associations yet increased cytoplasm or nucleoplasm signals (Fig. 2a, b, Extended Data Figs. 5e, f, 6 and Supplementary Table 4).

To reveal immediate effects in an inducible way, we sought to deplete the U1 snRNP component SNRNP70 by using an auxin-inducible degron (AID) system (SNRNP70^{AID}) in mESCs (Methods). Four-hour treatment with auxin led to a 90% depletion of SNRNP70 protein, which impaired the binding of U1 snRNA to its targeted lncRNAs, and attenuated splicing, but

had a minor effect on the expression of the lncRNAs analysed, and did not elicit apoptosis (Extended Data Fig. 5g–k). Consistently, total RNA-seq of auxin-treated SNRNP70^{AID} mESCs revealed 346 chromatin-downregulated lncRNAs (27%; $P < 0.05$), which substantially overlap with those affected by U1 AMOs and show increased signals in the cytoplasm and/or nucleoplasm fractions (Fig. 2, Extended Data Figs. 5f, l, m, 6 and Supplementary Table 5). The sets of chromatin-downregulated lncRNAs following U1 AMO and/or SNRNP70^{AID} treatment exhibit stronger U1 snRNA binding activities and higher expression levels compared with lncRNAs that are not downregulated, providing evidence that U1 snRNP promotes chromatin association of its target lncRNAs (Extended Data Fig. 5n).

We then investigated whether the U1 mechanism regulates the localization of mature transcripts after a lncRNA is made. Sequencing of polyadenylated RNA (polyA-seq) revealed that degradation of SNRNP70^{AID} led to roughly 23.6% of lncRNAs (295; $P < 0.05$) being downregulated on chromatin, which correlated significantly ($R = 0.49$, $P = 5.0 \times 10^{-19}$) with the change in total RNA-seq (Fig. 2a, b, Extended Data Figs. 6, 7a–c and Supplementary Table 6). To investigate whether newly synthesized polyA RNA shows similar changes, we pulse-labelled SNRNP70^{AID} mESCs with 4-thiol-uridine (4sU) and performed thiol-linked alkylation for metabolic sequencing (SLAM-seq²⁴). Chemical conversion of the newly incorporated 4sU into cytidine discriminates new transcripts from pre-existing transcripts (Extended Data Fig. 7d, e). Among 492 polyA lncRNAs with new transcripts detectable by SLAM-seq, 115 (23.4%) were downregulated on chromatin in auxin-treated mESCs, while only 3 were upregulated (Fig. 2c and Supplementary Table 7). New transcripts and all transcripts (new plus pre-existing) show highly correlated changes ($R = 0.81$, $P = 4.1 \times 10^{-117}$) upon SNRNP70^{AID} degradation (Extended Data Fig. 7f). Notably, both well spliced lncRNAs (for example, *Meg3* and *Rian*) and poorly spliced lncRNAs (such as *Lncenc1*, *Tsix* and *Pvt1*) show decreased chromatin binding after U1 inhibition (Extended Data Fig. 6). These findings rule out a kinetic effect due to delayed release of nascent or unspliced RNAs from their transcription sites that contributes mainly to chromatin retention.

Chromatin-downregulated lncRNAs with slightly longer transcript length exhibit similar distributions of total RNA signals across the gene body before and after U1 inhibition, and do not show higher decreases in expression compared with lncRNAs that are not downregulated (Extended Data Figs. 5n, 7g, h), arguing against globally premature

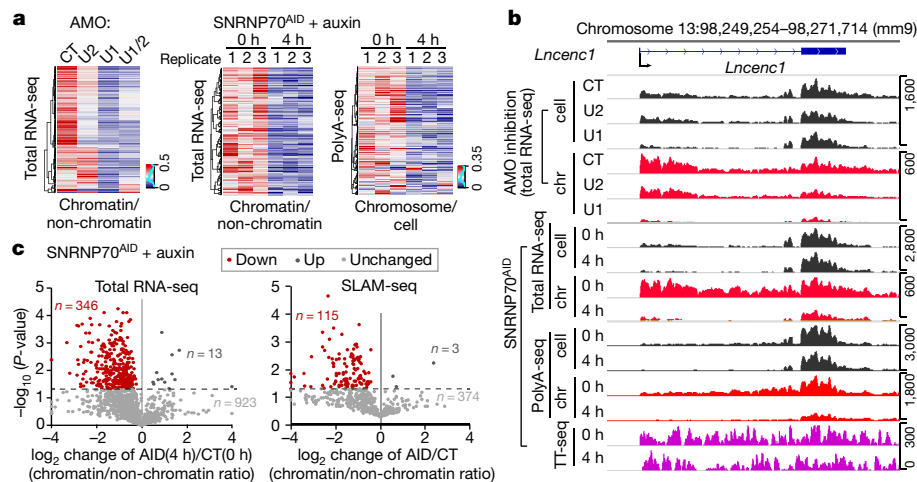


Fig. 2 | U1 snRNP regulates lncRNA-chromatin retention. **a**, Heat maps showing, left, the chromatin/non-chromatin ratio (where ‘non-chromatin’ refers to cytoplasm plus nucleoplasm) of 337 chromatin-downregulated lncRNAs in total RNA-seq upon AMO treatment; and right, the chromatin/non-chromatin ratio of 346 chromatin-downregulated lncRNAs in total RNA-seq and the chromatin/cell ratio (where ‘cell’ refers to the whole cell) in polyA-seq upon SNRNP70^{AID} degradation. See also Extended Data Fig. 5f. CT, no auxin. **b**, A genome-browser view of the *Lncenc1* locus; see also Extended Data Fig. 6g. In this panel, mm9 is a mouse

genome assembly. The top two sets of tracks show the total RNA-seq of whole-cell and chromatin fractions after treatments with control (CT), U2 or U1 AMOs. The lower three sets of tracks show total and polyA RNA-seq of whole-cell and chromatin fractions and TT-seq in SNRNP70^{AID} mESCs at 0 h or 4 h of auxin treatment. **c**, Volcano plots showing fold changes (log₂) of the chromatin/non-chromatin ratio of 1,282 lncRNAs identified by total RNA-seq or of 492 newly synthesized lncRNAs identified by SLAM-seq upon SNRNP70^{AID} depletion. AID, plus auxin. *P*-values were obtained by two-sided *t*-test for three biological replicates.

termination²⁵. To further assess potential transcriptional changes, we performed transient transcriptome sequencing of nascent transcripts (TT-seq) and RNA polymerase II (Pol II) chromatin immunoprecipitation followed by sequencing (ChIP-seq). Acute degradation of SNRNP70^{AID} did not alter the overall transcription rate and the distribution of paused and elongating Pol II across the genome (Extended Data Fig. 7i–k). Moreover, depletion of *EXOSC3*—an essential subunit of the RNA exosome—failed to rescue the observed decreases of chromatin-bound lncRNAs in SNRNP70^{AID} mESCs (Extended Data Fig. 7l). These combined analyses elucidate a direct effect of U1 snRNP in tethering both nascent and polyadenylated lncRNAs to chromatin, excluding the possibility that there are indirect consequences due to changes in transcription dynamics, RNA processing and decay on lncRNA–chromatin associations under our assay conditions of inhibiting U1 snRNP for short periods of time.

We detected substantial amounts of SNRNP70 and SNRPC on chromatin, and this association was sensitive to high-salt extraction (Extended Data Fig. 8a). Mass-spectrometry analysis of the SNRNP70 complexes identified proteins involved in transcription regulation, such as the Pol II large subunit POLR2A and elongation factors SPT5 and SPT6, besides splicing factors (Extended Data Fig. 8b, c and Supplementary Table 8). Using native chromatin extracts that were released by the nuclease benzonase—which degrades RNA and DNA—we confirmed that SNRNP70 captured phosphorylated Pol II (S2P, phosphorylated at serine 2) and SPT6 in wild-type but not auxin-treated SNRNP70^{AID} mESCs (Extended Data Fig. 8d). Reciprocal co-immunoprecipitation of phosphorylated Pol II (SSP and S2P), but not hypophosphorylated Pol II (8WG16), captured SNRNP70 and SNRPC (Fig. 3a). Thus, U1 snRNP binds transcriptionally engaged Pol II on chromatin, in a manner that is likely to be independent of RNA and DNA. Treatment of mESCs with the transcription inhibitors flavopiridol or triptolide reduced levels of chromatin-bound SNRNP70 and SNRPC proteins and also U1 snRNA (Fig. 3b and Extended Data Fig. 8a, e–g), consistent with a previous report of disrupted enrichment of U1 snRNA in the gene body of active genes by flavopiridol¹⁷ (Supplementary Discussion 2). Inhibition of transcription also led to reduced chromatin associations of analysed lncRNAs (Fig. 3b and Extended Data Fig. 8e). These results suggest a central role of the active Pol II machinery in promoting the chromatin association of U1 snRNP and its target lncRNAs.

We then investigated whether other spliceosome components might contribute to lncRNA–chromatin retention. Treatment of mESCs with the drug E7107, which specifically targets the U2 snRNP²⁶, also attenuated chromatin associations of analysed lncRNAs (Fig. 3b and Extended Data Fig. 8e). By contrast, inhibiting the recruitment of the U4/U6–U5 tri-snRNPs to the catalytically active spliceosome by using the drug isoginkgetin²⁷ or by depletion of the tri-snRNP components PRPF8 or SNRNP200 failed to have an effect (Fig. 3b and Extended Data Fig. 8e, h). We further inactivated U2 or both U1 and U2 (U1/2) snRNAs with AMOs. Inhibition of U2 snRNA alone had a much subtler effect, altering the localization of a subset of chromatin-bound lncRNAs, and inhibition of both U1/2 snRNAs did not result in a stronger effect than did inhibition of U1 snRNA alone (Fig. 2a, b and Extended Data Figs. 5a–f, 6). We posit that precatalytic recognition by U1 snRNP, and to a lesser degree by U2 snRNP, but not splicing per se primarily controls lncRNA–chromatin retention. Notably, for promoter-associated upstream antisense (ua) RNAs and enhancer (e)RNAs^{2–5}, inhibition of U2 snRNA as well as U1 snRNP also led to decreased chromatin but increased cytosolic signals (Extended Data Fig. 9).

Lastly, to reveal the biological significance of U1 mechanism, we focused on *Malat1*, which binds to thousands of genomic sites to regulate transcription, pre-mRNA splicing and nuclear architecture^{17,28}. Mature transcripts of *Malat1* are neither spliced nor polyadenylated, providing an ideal example to reveal how U1 snRNP affects its chromatin localization, independently of splicing and PAS-mediated RNA decay^{21–23}. U1 snRNA binds extensively to *Malat1* RNA, although it is not spliced, and inhibition of U1 snRNP led to decreased levels of *Malat1* on chromatin without severely altering its expression (Extended Data Figs. 2c, 5b and 6a). We performed *Malat1* chromatin isolation by RNA purification and sequencing (ChIRP-seq) and RNA fluorescence in situ hybridization (FISH). Rapid degradation of SNRNP70^{AID} caused drastically reduced *Malat1* binding at active genes across the genome, and abolished the punctate, speckle-like staining pattern of *Malat1*, whereas the localization of SC35 protein—a marker of nuclear speckles—was not affected (Fig. 3c, d and Extended Data Fig. 10a–e). Moreover, treatment with triptolide also abolished *Malat1* binding to its own and other genomic sites (Extended Data Fig. 10d), implying a Pol II transcription-dependent retargeting of *Malat1* to chromatin.

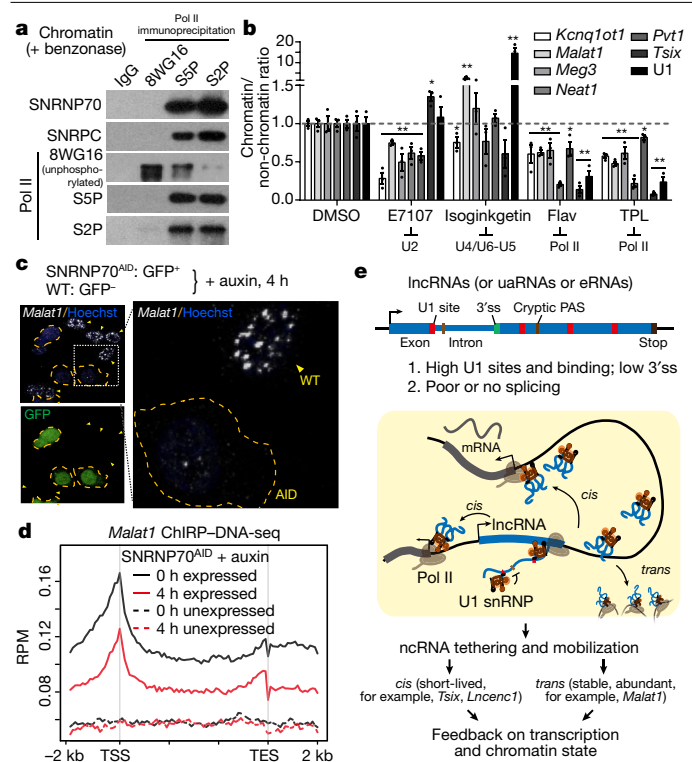


Fig. 3 | U1 tethers and mobilizes lncRNAs to chromatin by interacting with transcriptionally engaged Pol II. **a**, Co-immunoprecipitation and western blot analysis of chromatin fractions released by benzonase; $n = 3$ independent experiments. **b**, Relative chromatin/non-chromatin ratios of representative lncRNAs and U1 snRNA in various treatments. DMSO, dimethylsulfoxide (vehicle); Flav, flavopiridol; TPL, triptolide. Data are mean $\pm$ s.e.m. P -values were obtained by two-sided t -test for three biological replicates. $^*P < 0.05$, $^{**}P < 0.01$. **c**, FISH of *Malat1* RNA. SNRNP70^{AID} mESCs (GFP⁺, dashed circles) and wild-type mESCs (WT; GFP⁻, arrows) were mixed and treated with auxin for 4 h. A zoomed-in view of the boxed region is shown on the right. Hoechst staining indicates DNA. $n = 3$ independent experiments; the statistical summary is shown in Extended Data Fig. 10c. **d**, Meta-analysis of *Malat1* ChIRP-DNA-seq signals across the gene body of 10,675 expressed and 7,933 unexpressed genes. TSS, transcriptional start site. **e**, U1 snRNP mobilizes lncRNAs to chromatin during their synthesis and function. Top, after an lncRNA (or uaRNA or eRNA) is synthesized, it binds persistently to U1 snRNP owing to an imbalanced distribution of 5' and 3' splice sites. Bottom, this provides a way to mobilize lncRNA transcripts to nearby gene or regulatory sites within their chromatin neighbourhoods (for many cis-regulatory lncRNAs) or to distal chromatin regions (as exemplified by the lncRNA *Malat1*). Retargeting of lncRNAs by U1 snRNP to cis and/or trans genomic sites may be achieved in part through their interactions with engaged Pol II. Consequently, chromatin-bound lncRNAs may function as an RNA glue to hold U1 snRNP and the Pol II machinery at regulatory sites, creating a reservoir of regulatory factors for feedback regulation of transcription and chromatin state. For gel source data, see Supplementary Fig. 1.

In summary, unlike mRNAs, lncRNAs are enriched with U1-recognition sites but depleted of 3' splice sites (Fig. 3e). Dynamic interaction of U1 snRNP with transcriptionally engaged Pol II may provide a means of mobilizing U1 snRNP and its interacting lncRNA transcripts to cis and trans genomic sites for feedback regulation of transcription and chromatin state^{11–13}. This U1 model provides a parsimonious mechanism that is generally applicable to hundreds of noncoding transcripts, although other mechanisms—such as those involving repetitive sequences^{14–16,29}—may also exist to achieve lncRNA association with chromatin. This newly identified function of U1 snRNP adds to other findings that suggest a role for U1 beyond splicing, such as facilitating promoter directionality through the U1–PAS axis^{23,30}, and promoting transcriptomic integrity through U1 telescripting²⁵. It is possible that these mechanisms work

in concert to ensure the proper expression and function of lncRNAs on chromatin (Extended Data Fig. 10f).

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-020-2105-3>.

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Methods

Cell culture, transfection, and treatment

Mouse mESCs (46C) were maintained in complete ESC culture medium under standard ESC culture conditions as described³¹. HEK 293T and HeLa cells were maintained in standard eukaryote cell culture medium. Cell lines were not authenticated, and were routinely tested for mycoplasma contamination by polymerase chain reaction (PCR). Lipofectamine 3000 was used for transfection. The plasmids from the REL-seq pools or reporters were co-transfected with plasmids expressing the pBASE transposase. At 12–16 h after transfection, the medium was replaced with the appropriate medium containing 250 $\mu\text{g ml}^{-1}$ hygromycin B (Thermo Fisher) for 2 days; the cells were then transferred into a 10-cm plate for subsequent subcellular fractionation assays. For drug treatments, under culture conditions, cells were treated with 1 μM E7107 (an inhibitor of U2 snRNP²⁶, from Y. Shi's laboratory), with 1 μM flavopiridol (a transcription inhibitor³²; Selleck, catalogue number S1230), or with 1 μM triptolide (another transcription inhibitor³³; Abcam, ab120720) for 1 h, or with 100 μM isoginkgetin (an inhibitor of U4/6-U5 snRNP²⁷; MedChemExpress, HY-N2117) for 2 h.

Construction of the SNRNP70^{AID} cell line

To construct the auxin-inducible degron (AID)³⁴ cell line for SNRNP70, we co-transfected into mESCs an exogenous SNRNP70 expression construct with its amino terminus fused to 3 $\times$ FLAG–biotin–AID tags in the *PiggyBac* vector (PB–3FB–AID–SNRNP70; hygromycin-resistant), together with E3 ligase expression plasmids for the AID system (PB–Tir1; neomycin-resistant) and plasmids expressing the transposase pBASE. After selection with drugs for five days, endogenous SNRNP70 was deleted through the CRISPR/Cas9 system and single clones were picked as described¹⁰. Clones with homozygous deletion of endogenous SNRNP70 and rapid removal of exogenous AID-tagged SNRNP70 were used for downstream experiments. The auxin analogue indole-3-acetic acid (IAA) was used to induce the degradation of AID-tagged SNRNP70 protein.

Subcellular fractionation assay

Subcellular fractionation was performed as described with the following modifications³⁵. Cytoplasmic buffer containing 0.05% or 0.15% NP-40 was used to isolate the cytoplasmic fraction of mESC or 293T cells respectively. The whole of each subcellular fraction was spiked and mixed thoroughly with 5 μl of 0.1 ng μl^{-1} spike-in RNA (containing two *in vitro* transcribed RNAs, *lacZ* and *mCherry*; for polyA RNA-seq or SLAM-seq, ERCC spike-in (Thermo Fisher) was used). For total RNA-seq, ribosomal RNAs were removed using Ribo-Zero rRNA removal kits (Epicentre); for polyA RNA-seq, polyA RNAs were isolated using Dynabeads mRNA purification kits (Thermo Fisher). RNA-seq libraries were constructed using NEBNext Ultra II directional RNA library prep kits (NEB). For reverse transcription with quantitative PCR (RT–qPCR) or RNA-seq data analysis, the ratio of each fraction versus total input was calculated by normalizing to the spiked transcript. For protein analysis, the same ratio of lysate from the different fractions versus input was used for western blotting.

We also performed a subcellular fractionation with sequentially increasing salt concentrations as described³⁶. Briefly, isolated nuclei were washed for 10 min with soluble nuclear buffer (1 mM EDTA, 0.2 mM EGTA), then washed sequentially with soluble nuclear buffer supplemented with 75 mM, 150 mM, 300 mM or 600 mM NaCl. The pellet was then digested with benzonase nuclease, and the supernatant was harvested after spinning at 14,000 r.p.m. for 15 min. The supernatant of each step was collected, and the same portion of each collected sample was used for further western blotting.

REL-seq and mutREL-seq

DNA fragments of candidate genes were obtained from PCR of genomic DNA (*Malat1*, *Neat1*, *NXFI-IR* (where IR is the retained intron 10 of *NXFI*),

NR_028425 and the 3' untranslated repeat (UTR) region of *NCL*) or complementary DNA (*ACTB*, *NANOG*, 5'-UTR and the coding-sequence region of *NCL*). For *Xist*, the pCMV-*Xist* plasmid (which also contains a roughly 2-kilobase region of the last exon of *Tsix*) was ordered from Addgene and used for REL-seq³⁷. DNA samples were sonicated to obtain a mixture of short DNA fragments of the expected size (Extended Data Fig. 1c and Supplementary Table 1). For mutREL-seq, random mutagenesis was achieved through error-prone PCR as described³⁸.

The short DNA fragments were end-repaired and adenylated using a DNA library construction kit according to the manufacturer's instruction (NEB). In-house-designed 'Y-shaped' adaptors (Supplementary Table 9) were then ligated to the prepared fragments, and the ligation products were further size-selected by agarose-gel purification. Note that the adaptor can be ligated in both sense and antisense directions. The reverse-complement inserts, which were generated as an insertion by-product, serve as an internal control for the sense strand. The purified products were amplified, digested by *AscI* and *NotI*, and ligated with *AscI*- and *NotI*-digested 5AI, 3AI (which refer to 3' or 5' of the *ACTB* intron) or GFP reporter vectors. The products were further purified by ethanol precipitation and the pellet was dissolved in 1 μl of water. See Supplementary Note 1 for a description of the three reporter vectors. The purified ligation products were transformed into electrocompetent cells (Takara) through electroporation and plated evenly on two 15-cm agar plates. After overnight growth, the cells were harvested by scraping, and the plasmids were then extracted and co-transfected with pBASE into mESCs or 293T cells as described above. These two cell lines were chosen as they can be efficiently transfected, and they represent pluripotent and fully differentiated cells in mice and humans, respectively. After 2 days of drug selection, cells were plated onto a 10-cm plate, and subcellular fractionation was performed and RNA was extracted from the different fractions using TRIzol reagent. RNAs were further treated with DNase I for 20 min to remove residual DNA, and reverse-transcribed with SuperScript III reverse transcriptase (Thermo Fisher). A specific reverse-transcription primer that binds downstream of the insertion site was used for reverse transcription (Supplementary Table 9). Reverse transcription was performed at 50 °C for 40 min and at 55 °C for another 20 min. Ten units of exonuclease I were added and incubated at 37 °C for 20 min to remove the free reverse-transcription primers. The reaction was stopped by heating the sample at 95 °C for 10 min to inactivate all of the enzymes. Then, 0.5 μl of 10 mg ml^{-1} RNase A was added and incubation was carried out at 37 °C for 20 min to degrade the RNA. Complementary DNAs were purified through ethanol precipitation and used as templates for PCR and library construction.

PCR was performed using primers that bind on either side of the inserted fragments. For the 5AI or 3AI reporter, one primer specifically targeted the exon junction site of the spliced *ACTB* intron 3, while the other primer targeted the adaptor sequence. For the GFP reporter, both of the primers were designed to target the adaptor sequences (Supplementary Table 9). The PCR products were then purified using 1 $\times$ Ampure XP beads, and libraries for the different samples were constructed using NEBNext Ultra II DNA library prep kits (NEB, E7645).

RNA pull-down assay

RNA pull-down assays were performed as reported with some modifications³⁹. Briefly, biotinylated *NXFI*-enChr sense or antisense (*NXFI*-enChr-as) RNA was obtained by *in vitro* transcription with biotin-UTP incorporation. We denatured 2 μg of purified biotinylated RNA at 90 °C for 2 min, and then transferred it onto ice for 2 min. We added a one-quarter volume of 5 $\times$ RNA structure buffer (final concentration 10 mM Tris pH 7, 0.1 M KCl, 10 mM MgCl₂) and incubated the mixture at room temperature for 20 min. The folded RNA was mixed with the precleared mESC nuclear extract, and incubated at room temperature for 1 h. Prewashed streptavidin beads were added and incubated at room temperature for another 2 h. The beads were washed five times with RIP wash buffer (200 mM KCl, 25 mM Tris pH 7.4), then eluted with

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nuclear lysis buffer (50 mM Tris-Cl, pH 7.5, 10 mM EDTA, 1% SDS) at room temperature for 2 min. A one-fifth volume of 6× SDS sample buffer was added and boiled for 5 min at 95 °C, and the protein sample was used for western blotting or prepared for mass-spectrometry analysis.

AMO inhibition

Treatment with AMOs for U1, U2 or U1/2 snRNA inhibition was performed as described with modifications^{25,30,40,41}. Briefly, roughly 2.5×10^6 mESCs were nucleofected with various AMOs by electroporation using a Nucleofector (Amaxa) and then immediately plated in a six-well plate. For each treatment with scrambled control (CT), U1 or U2 AMOs, the AMO concentration was 7.5 nmol per 100 µl (75 µM) per nucleofection. For inhibition of both U1/2, 5 nmol per 100 µl of U1 and U2 AMOs (50 µM each) were co-transfected. AMO sequences are listed in Supplementary Table 9. After 2 h of AMO nucleofection, cells were harvested for RNA-seq (whole cells) or subcellular fractionation for downstream experiments and analyses. Note that the AMO-treated mESCs were nicely attached to the culture plate and no obvious morphological changes were observed at the end of the 2-h treatment time. However, mESCs appear to be more sensitive to U1 and/or U2 AMOs than were cell lines used in previous studies, where AMO treatment was performed for 8 h (refs. ^{25,30,40,41}). After 4-h treatment with U1, U2 or U1/2 AMOs, mESCs started to become detached from the culture plate, indicating cell death. Extensive cell death was observed at 8 h of AMO nucleofection. Therefore, we chose to analyse the immediate effects of U1 and/or U1 inhibition on RNA subcellular localization after 2 h of treatment instead of 8 h. Short-term treatments appear to be suitable for studying lncRNA and ncRNA transcripts with relatively short half-lives⁴².

Co-immunoprecipitation

Co-immunoprecipitation was performed as described with some modifications⁴³. mESC nuclei were isolated using hypotonic buffer and washed with benzonase digestion buffer (50 mM HEPES, pH 7.5, 200 mM NaCl, 5 mM MgCl₂, 0.1 mM Na₃VO₄, 0.1% NP-40, 10% glycerol) at 4 °C for 30 min. The pellet was resuspended with benzonase digestion buffer containing 1 µl benzonase nuclease (Sigma, E1014) and rotated at 4 °C for 30 min to isolate the chromatin extract. For antibody immunoprecipitation, we added 2 µg of antibody to each sample and used protein G magnetic beads (Thermo Fisher) for the pull-down. FLAG M2 magnetic beads (Sigma) were used for FLAG immunoprecipitation.

For co-immunoprecipitation coupled with stable-isotope labelling by amino acids in cell culture followed by mass spectrometry (colP-SILAC), wild-type mESCs were cultured with heavy SILAC media and mixed with equal amounts of SNRNP70^{AID} mESCs (FLAG-tagged) cultured in light SILAC media. The co-immunoprecipitation procedure was as described above, except that pull-down was performed with FLAG beads and the proteins were eluted with FLAG peptide. The eluate was concentrated through acetone precipitation. Protein identity and relative enrichment were determined by mass spectrometry. Proteins with scores larger than 20 and light/heavy ratios larger than 2 were defined as SNRNP70-interacting proteins (Supplementary Table 8).

SLAM-seq of subcellular fractionation samples

The SLAM-seq protocol was modified from a previous report²⁴. Briefly, SNRNP70^{AID} mESCs were treated with auxin for 2 h, then allowed to incorporate 4sU (300 mM final) for 3 h in the presence of auxin. The purpose of this 3-h 4sU labelling is to make sure that most of the new transcripts are fully transcribed, processed and localized. SNRNP70^{AID} mESCs treated with 4sU but without auxin were used as controls. Subcellular fractionation was performed as described above and RNAs from each fraction were extracted. The RNAs were further treated with chemical reactions as described⁴⁴ and 3'-end mRNA-seq libraries were constructed using a commercially available kit (QuantSeq 3'mRNA-Seq library prep kit, Lexogen). Deep sequencing was performed and data

were analysed using SLAM-DUNK⁴⁵, with the last exon of lncRNAs as the reference. The remaining parameters were left as defaults.

TT-seq

The TT-seq protocol was modified from a previous report⁴⁶. Auxin-treated or untreated SNRNP70^{AID} mESCs were allowed to incorporate 4sU (500 mM final) for 10 min; cells were then harvested and RNAs were extracted by TRIzol. Labelled RNAs were further biotinylated as described⁴⁷ and fragmented with 0.1× NaOH on ice for 10 min. The fragmented RNAs were subjected to purification as described⁴⁷. RNA-seq library construction, deep sequencing and data analysis were further performed as described above and below.

RNA FISH

Malat1 single-molecule inexpensive FISH (smRNA FISH) was performed as described⁴⁸. Immunofluorescence was carried out according to the protocol provided by Cell Signaling Technology. To reduce variations arising from experimental or imaging procedures, we constructed a SNRNP70^{AID} mESC line with stable integration of a *GFP* gene (*GFP*⁺). SNRNP70^{AID}-*GFP*⁺ mESCs were mixed with wild-type *GFP*-negative mESCs (*GFP*⁻) before plating for RNA FISH or immunofluorescence. The probes used for *Malat1* FISH are listed in Supplementary Table 9.

ChIRP-seq

Auxin-treated or untreated SNRNP70^{AID} cells were crosslinked using 2 mM dithiobis succinimidyl propionate (DSP) for 30 min, followed by 15 min of crosslinking with 3.7% formaldehyde. The crosslinked cells were first partially digested with 12 U ml⁻¹ DNase I at 37 °C for 10 min, then sonicated at 25% amplitude for 30 s. Subsequent procedures were performed as described^{10,49}. The probes used for *Malat1* ChIRP are listed in Supplementary Table 9. For ChIRP RNA, auxin-treated or untreated SNRNP70^{AID} mESCs were crosslinked with 2% formaldehyde for 10 min. Subsequent procedures were performed as described¹⁰.

REL-seq data analysis

Sequences corresponding to vectors and adaptors were removed from the REL-seq sequencing data using BEDTools⁵⁰. The remaining 'clean' reads were mapped to the mouse genome (mm9) through TopHat⁵¹. To compare read densities, we divided the genomic regions of candidate genes into multiple bins, each ten base pairs in length. For each sample, the number of reads located in each bin was counted and further normalized by the total reads mapped to those bins. The fold changes were calculated by dividing the read density of the chromatin fraction by the read density of the cytoplasm fraction. To identify the chromatin-enriched regions, we calculated the fold changes of the different samples, and carried out a *t*-test to compare the fold change of each sample with the median fold change of the corresponding sample. Only a bin with a *t*-test *P*-value of less than 0.05 and a fold change greater than 1.5 was recognized as a bin with significant chromatin enrichment. Neighbouring bins, within a distance of 50 nucleotides, were merged as a region with significant chromatin enrichment. As most of the inserts were much larger than 50 nucleotides, merged inserts with a length of less than 50 nucleotides were discarded. Because we used very stringent selection criteria across all REL-seq data sets generated with different libraries in three AI (*ACTB* intron) and *GFP* reporter vectors in both human and mouse cell lines, the enChrs we identified may underrepresent the total number of regulatory RNA signals for chromatin association in the cell.

To identify inserts that were enriched in different fractions, we reconstructed inserts by identifying their ends through the mapping of paired-end reads. The abundance of different inserts in the different fractions was counted, and fold changes of different clones were calculated by dividing the insert abundance in the chromatin fraction by the insert abundance in the cytoplasm fraction. Only inserts with a maximum read number larger than ten and a minimum no less than one

were used for the analysis. The cutoff of fold changes for chromatin-enriched inserts is annotated in the relevant figure legends.

For *NXF1*-enChr mutREL-seq, 'clean' reads were mapped through Novoalign onto a reference sequence, built in-house, which contained only the sequence of the *NXF1*-enChr region. The different mutations at each position were counted. The cytoplasmic/chromatin ratio of each mutation was calculated by dividing the normalized reads (normalized by wild-type reads) in the cytoplasmic fraction by that in the chromatin fraction.

RNA-seq data analysis

RNA-seq data from fractionation of human K562 cells were downloaded from the ENCODE project (<https://www.encodeproject.org>)^{52,53}. Alignments of RNA-seq data to human genome assembly hg19 were performed using Tophat v2.0.10 (ref.⁵⁴). Fragments per kilobase of exon model per million mapped reads (FPKM) were calculated using Cufflink 2.1.1 to represent expression levels of transcripts. Gencode v19 was used as the human gene annotation. Similarly, RNA-seq data from subcellular fractionation of mouse cells were mapped to mouse genome assembly mm10 (for comparisons and correlation analyses) or mm9 (for tracks showing sequencing signals), and the corresponding FPKM were calculated with the Gencode vM9 annotation. For data analysis of strand-specific RNA-seq data, the strand information of each mapped read was first converted into absolute strand information relative to the genome; then the FPKM was calculated through Cufflinks 2.1.1 using stranded RNA-seq parameters.

For RNA-seq analysis of whole-cell and subcellular fractionated samples, raw reads of mouse subcellular fractionation RNA-seq data were mapped to mouse genome assembly mm9 for sequencing signal analysis tracks or mm10 for FPKM calculations. For the analysis of relative gene abundance in different subcellular fractions, we constructed an in-house reference file by combining the genome assembly mm10 with our in-house spike-in sequence (*lacZ* and *mCherry*). The RNA-seq data of each fraction were mapped to the in-house reference. The FPKM value of each gene was calculated (with Gencode vM9 annotation), and the ratio of reads that mapped to spike-in was also calculated ((reads mapped to spike-in)/(total mappable reads)). The FPKM value of each gene was further normalized by the corresponding ratio of spike-in, and the ratio of each fraction was calculated by dividing the normalized FPKM value of each fraction by the sum of all three fractions (cytosol plus nucleus plus chromatin). The sequencing signal track of subcellular fractionation samples was also normalized by the ratio of spike-in. lncRNAs with a minimum FPKM of more than 0 and a maximum FPKM in all samples of more than 1 were chosen as mESC-expressed lncRNAs. To identify U1 snRNP-regulated lncRNAs, we first normalized the RNA signals in each compartment to spike-in RNA controls and then calculated relative enrichments on chromatin (chromatin/non-chromatin ratio) by comparing normalized RNA signals in the chromatin to the non-chromatin (cytoplasm plus nucleoplasm) fractions. Only those lncRNAs that showed greater decreases in the chromatin fraction than in the nucleoplasm and cytoplasm fractions (a chromatin/non-chromatin ratio of less than one; $P < 0.05$) upon U1 snRNP depletion were selected as U1 snRNP-regulated lncRNAs. In addition, we also compared lncRNA abundance in each of the chromatin, cytoplasm and nucleoplasm fractions to the total amount of lncRNA in all fractions, depicted as 'chromatin/total', 'cytoplasm/total' and 'nucleus/total'.

We used the following published data sets in our analysis: total RNA-seq of whole cells and the chromatin fraction of K562 cells, GSE30567; U1 and *Malat1* RAP in mESCs¹⁷, GSE55914; Pol II NTD ChIP-exo-seq in mESCs⁵⁵, GSE64825; Pol II 8WG16 ChIP-seq in mESCs⁵⁶, GSE49847; Ser 2 and Ser 5 phosphorylated Pol II ChIP-seq in mESCs⁵⁷, GSE112114.

Correlation analysis

We used two methods—bioinformatics prediction and U1 RAP-RNA—to analyse the U1-recognition site. Prediction of U1 snRNP recognition

sites was performed as described³⁰. On the basis of their maximum entropy scores, we categorized the predicted U1 snRNP recognition sites into predicted strong (top 50% of sites) and medium (bottom 25% to 50% of sites). We calculated the number of U1-recognition sites in the intron and exon regions for each transcript. The average density of predicted U1-recognition sites was calculated by dividing the total number of predicted U1-recognition sites in all exons (or all introns, and so on) by the total length of all exons (or all introns and so on). Only transcripts with a total exon length (or intron length) greater than 1 kilobase were chosen for further comparison of the density of predicted U1-recognition sites in lncRNAs and mRNAs.

Only genes with FPKM values of more than 1 were used to compare mRNAs with different chromatin-association tendencies and lncRNAs. As RNA-seq data of mouse subcellular fractions are unstranded, lncRNA transcripts that overlapped with coding genes were discarded. Transcripts that are entirely located in repeat elements were also discarded. The fold enrichment of each transcript was calculated by dividing the FPKM in chromatin RNA-seq data by the FPKM of the respective total RNA-seq data. For Extended Data Fig. 4e, we used FPKM values of transcripts (isoforms) rather than genes for analysis, as different isoforms of the same gene may show a different chromatin-association tendency.

For U1 RAP-RNA analysis, we obtained U1 RAP-RNA data from ref.¹⁷, which used formaldehyde crosslinking to obtain the U1 snRNA interacting RNAs. We first calculated the number of reads in the intron and exon region for each transcript. Only those transcripts containing at least ten reads in each sample were kept for further analysis. To calculate the U1 enrichment for the intron and exon region of a target RNA, we divided the reads mapping to the target exon or intron of each transcript in RAP-RNA by the respective input with normalization of sequencing depth. The fold enrichment of the U1 RAP-RNA signal in lncRNAs and coding genes was calculated and compared. The *t*-test (two-sided) was used to calculate the significance of the difference between two groups.

Genome-wide prediction of 3' splice sites

To identify the genome-wide distribution of predicted 3'ss, we generated a fasta file containing all 4,096 possible arrangements of eight-nucleotide motifs ending with 'AG'. Those motifs were mapped to the mouse or human genome through bowtie-1.0.0 to obtain the genomic distribution of each eight-nucleotide motif. Each mapped coordinate plus 12-nucleotide upstream and 3-nucleotide downstream (23 nucleotides in total) was used to calculate the 3'ss score (by MaxEntScan⁵⁸). Each coordinate with a score larger than 8.0 was recognized as a predicted 3' splice site (the median scores of Gencode vM9 annotated 3' splice sites were 8.7 in coding genes and 8.0 in lncRNAs).

Other bioinformatics analyses

The secondary structures of the wild-type and mutated *NXF1*-enChr regions were predicted from the Vienna RNA website with default parameters^{59,60}. The analysed REL-seq tracks and other tracks are shown in Integrative Genomics Viewer (IGV)⁶¹. Metaplots were drawn using ngs.plot⁶².

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this paper.

Data availability

All sequencing data are available in the Sequence Read Archive database under accession numbers SRP214639 and SRP125289. For gel source data, see Supplementary Fig. 1. Source data for Fig. 2a–c are provided with the paper (Supplementary Tables 4–7). All other data are available from the corresponding author upon reasonable request. Sequencing data have been deposited in the Gene Expression Omnibus under accession numbers GSE107131 and GSE134287.

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Author contributions X.S. and Y.Y. conceived the project and wrote the manuscript. Y.Y. performed most experiments and bioinformatics analyses. X.Z. and W.S. helped with subcellular fractionation; Y.H. and L.C. helped to construct SNRNP70^{40D} mESCs; J.Y.L. and Y.X. helped with correlation analysis of U1-RNA localization; P.L. carried out prediction of U1-recognition sites under the guidance of Q.C.Z.; G.S. provided initial batches of AMOs; B.T. helped with 3' ss analysis.

Competing interests The authors declare no competing interests.

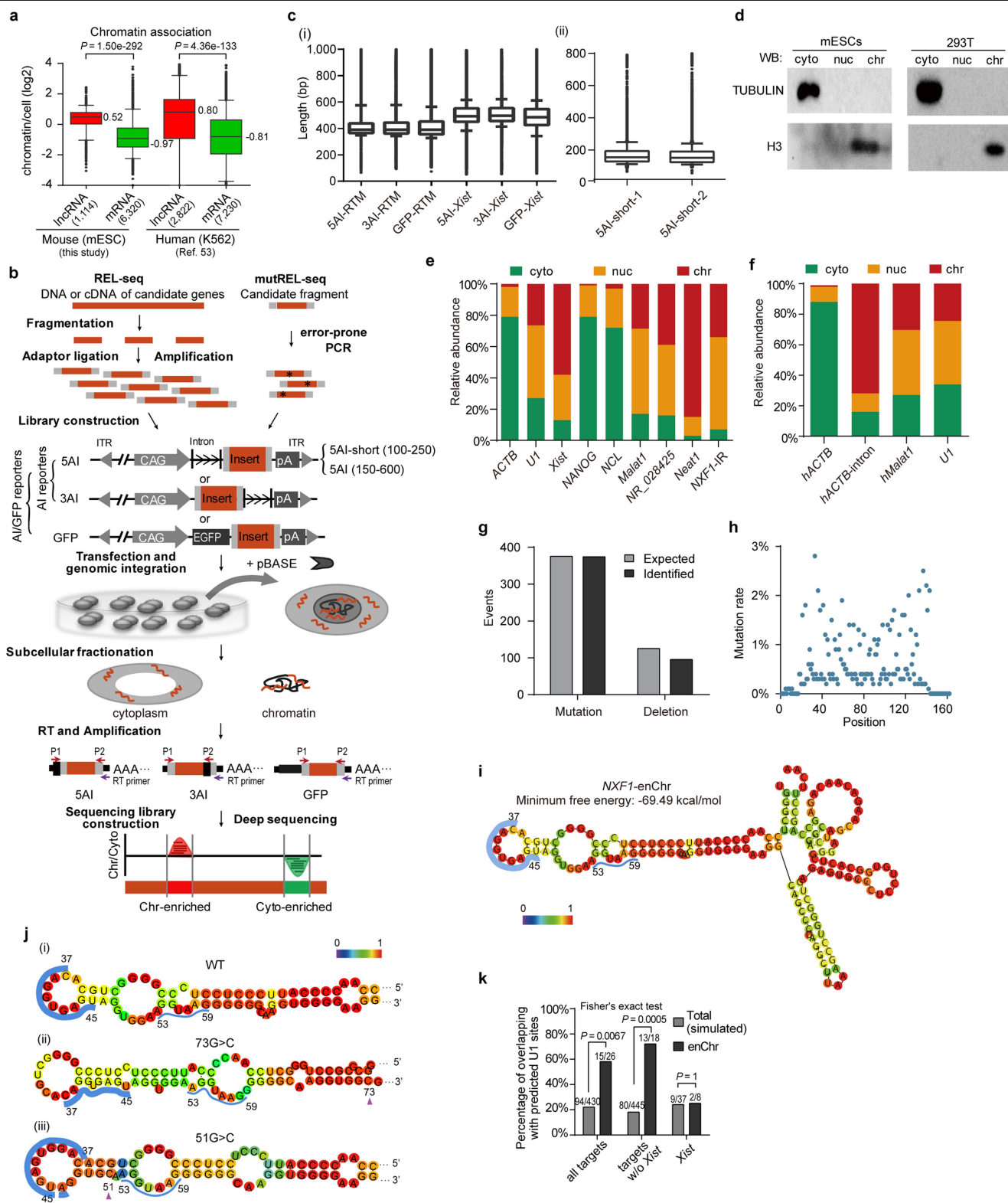
Additional information

Supplementary information is available for this paper at <https://doi.org/10.1038/s41586-020-2105-3>.

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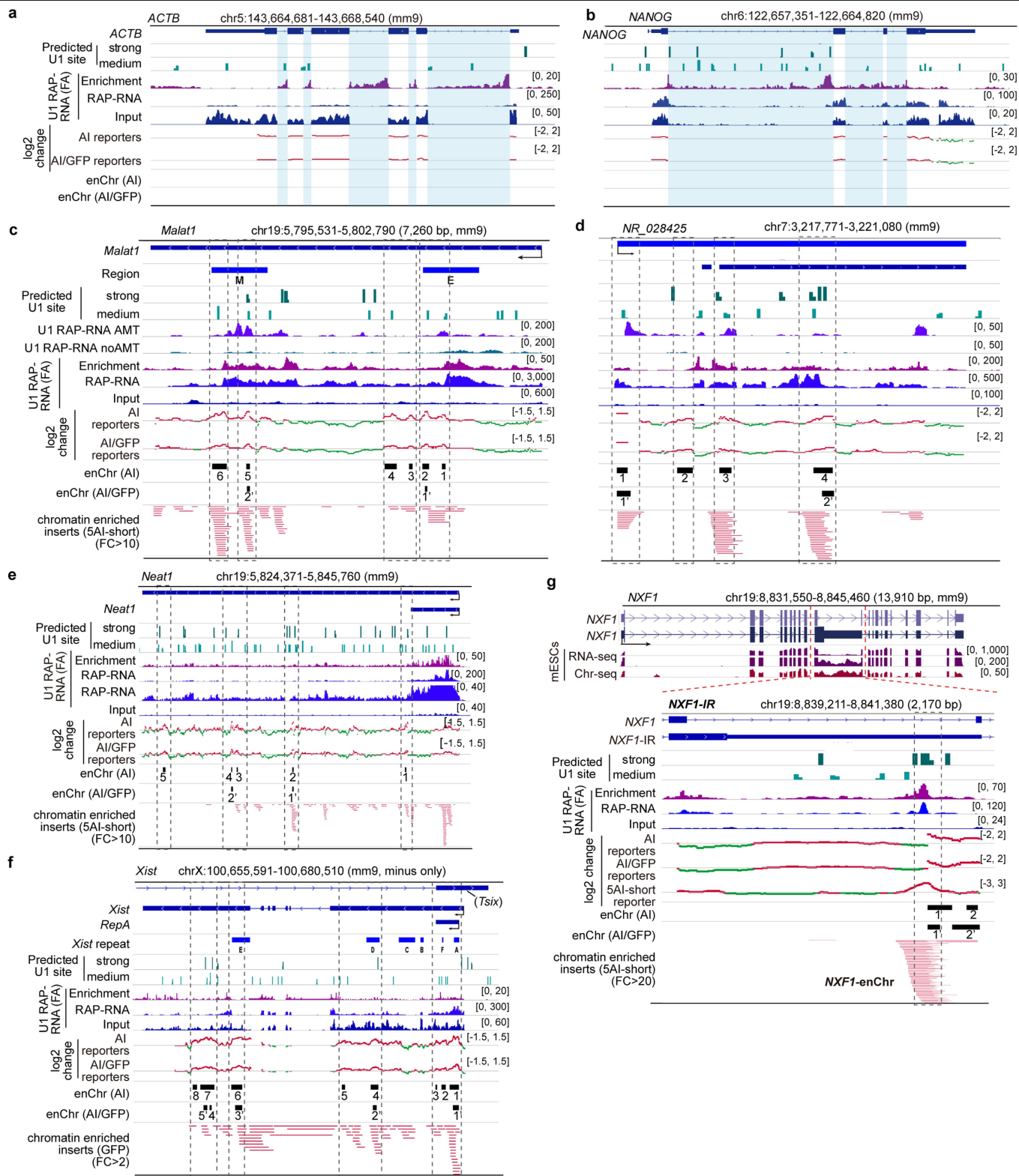
Extended Data Fig. 1 | See next page for caption.

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Extended Data Fig. 1 | REL-seq and mutREL-seq for identification of *cis* elements that contribute to the subcellular localization of RNA.

a, Comparisons of chromatin enrichment between lncRNAs and protein-coding mRNA genes in human and mouse. We fractionated chromatin-bound RNA in mESCs and compared the RNA-seq profiles of chromatin RNA to total RNA from whole cells. We also analysed a previously published RNA-seq data set (chromatin and cell) from human K562 cells⁵³. lncRNAs with FPKM values of greater than 1 and protein-coding genes with FPKM values of greater than 5 were used for the analysis. Data for samples connected by brackets were compared with two-sided *t*-tests. Consistent with previous reports^{6,8}, lncRNAs as a class are significantly enriched in the chromatin fraction in both mouse and human cells. Box plots show 5th, 25th, 50th, 75th and 95th percentiles, with median values labelled beside the box plots and sample sizes ($n \geq 1,114$) labelled on the x axis. **b**, Detailed pipelines for REL-seq and mutREL-seq. For REL-seq, DNA fragments from candidate genes are randomly fragmented, ligated with adaptors, amplified, and inserted into three types of reporters in the *PiggyBac* transposon vector (including 5AI, 3AI and GFP reporters). The reporters are co-transfected with *PiggyBac* transposases (pBASE) and stably integrated into the genome. RNA from the cytoplasm (cyto) and chromatin (chr) fractions is reverse-transcribed (RT) with primers indicated by purple arrows, and amplified with primers P1 and P2 for subsequent high-throughput sequencing. Sequences that are enriched in different subcellular fractions are identified by comparing the read intensities or insert abundance in the chromatin fraction with that of the cytoplasmic fraction. For mutREL-seq, a candidate fragment (*NXFI-enChr*) is randomly mutagenized through error-prone PCR, and the products are further inserted into the 5AI reporter vector and subjected to downstream procedures similar to those described for REL-seq. Asterisks represent mutation sites. ITR, inverted terminal repeat sequences of the *PiggyBac* transposon system. See also Supplementary Note 1. **c**, Box plots showing the length of inserts of RTM (RNA transcript mixture, without *Xist*) and *Xist* REL-seq libraries in 5AI, 3AI and GFP reporters (i) and 5AI-short reporters (ii). Box plots show 5th, 25th, 50th, 75th and 95th percentiles.

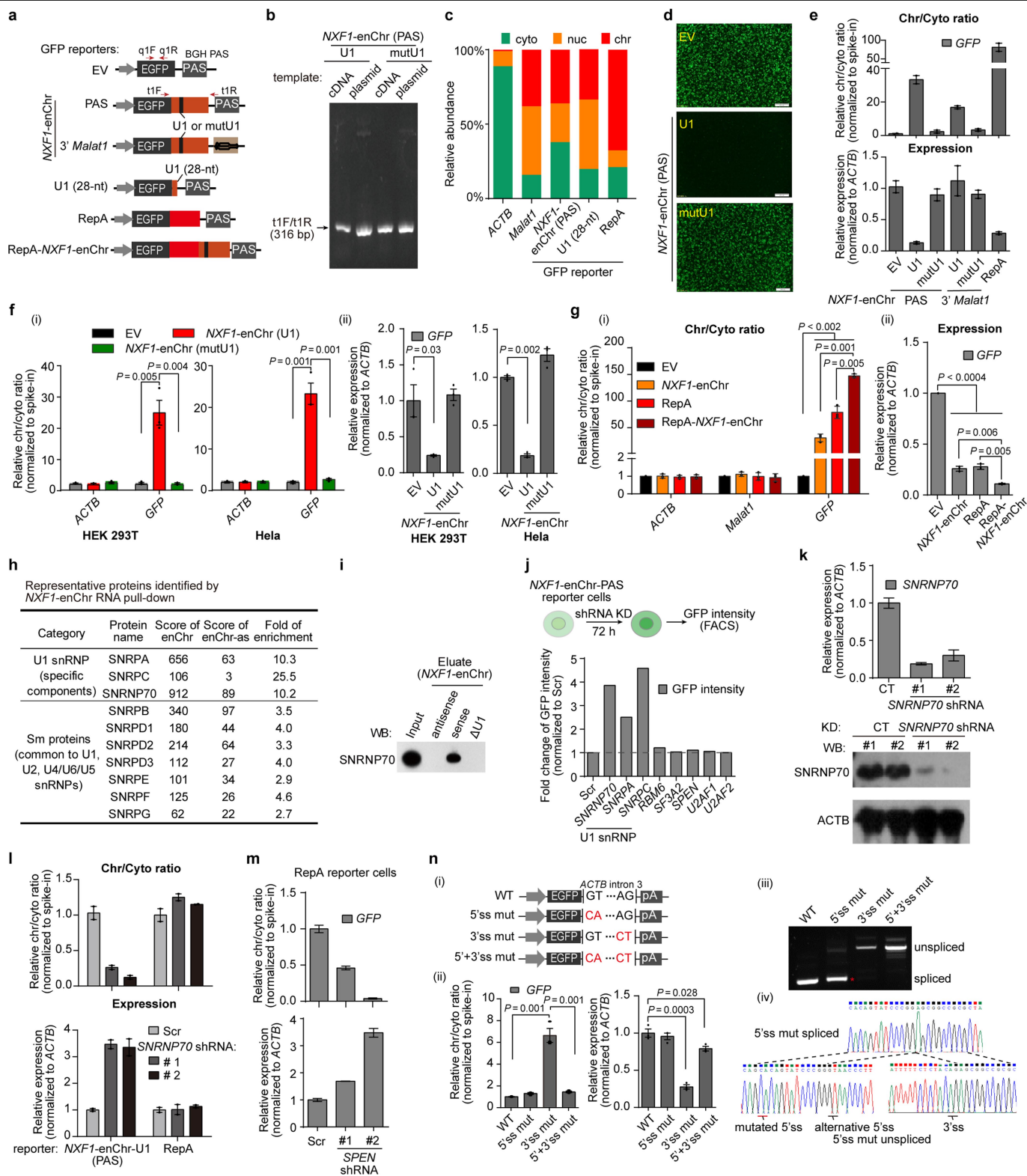
In i, $n = 20,000$ randomly selected inserts for each group. In ii, $n = 19,748$ for 5AI-short-1; $n = 15,091$ for 5AI-short-2. **d**, Western blot analysis of marker proteins in subcellular fractions of mESCs or HEK 293T cells. Tubulin and histone H3 are used as marker proteins for the cytoplasmic and chromatin fractions, respectively. $n = 3$ independent experiments. **e, f**, RT-qPCR analysis of the relative abundance of marker genes and candidate genes for REL-seq in subcellular fractions of mESCs (**e**) or human HEK 293T cells (**f**). *ACTB* and *Xist* are used as markers of mouse cytoplasmic and chromatin fractions, respectively; *hACTB* and *hACTB*-intron (intronic region of *hACTB*) are used as markers of human cytoplasmic and chromatin fractions, respectively. **g**, Numbers of mutation or deletion events expected and identified by mutREL-seq. A total of 469 mutation events, including 374 mutations (coverage 99.7%) and 85 deletions (coverage 68%), were identified over the 125-base-pair (bp) length of the *NXFI-enChr* DNA, indicating saturated mutagenesis. **h**, Analysis of the mutation rate. PCR mutations are spread across the *NXFI-enChr* sequence, ruling out a PCR bias towards the core 7-nucleotide mutations at positions 37–45. Notably, the two binding sites for PCR primers at the 5' (1–18-bp) and 3' (144–162-bp) ends of *NXFI-enChr* were less likely to be mutated, compared to the middle region with an average of 0.2%–3% mutation rate at each nucleotide position. This excludes the possibility that the sequences were misread. **i**, Predicted secondary structure of *NXFI-enChr* RNA. The U1-recognition site at positions 37–45 is highlighted with a thick blue line. A weak U1-recognition site at positions 53–59 is highlighted with a thin blue line. The coloured bar represents the probability of base-pairing or being unpaired (red, high probability of pairing (or lack of pairing); blue, low probability). **j**, Comparison of predicted secondary structures of the wild-type (i) and mutant (ii, iii) *NXFI-enChr* (partial sequences are shown). Purple triangles (ii, iii) highlight mutation sites. See also Supplementary Note 2. **k**, The percentage of enChrs or all target sequences ('total' = (sum of length)/(median length of enChr)) used for REL-seq analysis that overlap with predicted U1-recognition sites. *P*-values (one-sided Fisher's exact test) and sample sizes are shown at the top.



Extended Data Fig.2 | See next page for caption.

Extended Data Fig. 2 | Genome-browser views of representative genes showing REL-seq results. The enChrs (shown as thick black bars, also highlighted by dashed boxes) indicate regions with significant chromatin enrichment that were identified by AI/GFP or AI reporter screens ($P < 0.05$, fold change greater than 1.5). Representative chromatin-enriched inserts from respective reporters (SAI-short or GFP) are shown at the bottom of each panel. **a, b**, No enChrs were identified by REL-seq in the cytoplasm-localized protein-coding *ACTB* (**a**) and *NANOG* (**b**) transcripts. In these transcripts, U1 signals are mainly confined to the intronic regions and appear to be depleted in exons. Scales are shown in square brackets at the right of each track. **c–g**, REL-seq identified multiple enChrs in mouse *Malat1* (**c**), *NR_028425* (**d**), *Neat1* (**e**), *Xist* (**f**), and *NXF1-IR* (**g**) transcripts. In **c**, two mouse regions homologous to regions E and M in human *Malat1* are shown with thick blue lines. In **f**, only minus-strand tracks are shown. The plus-strand tracks for *Tsix*, which is transcribed in the antisense direction to *Xist* RNA, are shown in Extended Data Fig. 6c. The locations of different repeats in *Xist* are shown as thick blue lines. In **g**, multiple

strong U1-recognition sites are clustered in the 162-nucleotide *NXF1*-enChr (highlighted by dashed boxes) and nearby sequences. U1 snRNP also binds strongly (roughly 70-fold maximal enrichment versus the input) to this region, with peak signals that are centred at the 7-nucleotide U1 motif revealed by mutREL-seq (Fig. 1a), providing evidence for direct interactions of *NXF1*-enChr RNA and U1 snRNP in vivo. The predicted strong or medium U1-recognition sites are shown underneath the *NXF1* gene annotation. RNA-seq signals of U1 RAP-RNA with formaldehyde (FA) or 4'-aminomethyltrioxalen (AMT) crosslinking and respective input controls are also shown. AMT generates interstrand crosslinks between uridine bases to detect the direct RNA–RNA interactions of highly expressed transcripts¹⁷. FA stabilizes both direct and indirect interactions of proteins and nucleotides. Also shown is the average fold change ($\log_2$) of the read intensity of the chromatin fraction compared with that of the cytoplasmic fraction in inserts from AI/GFP reporters or AI reporters. Red lines represent chromatin enrichment, while green lines represent chromatin depletion, of signals.

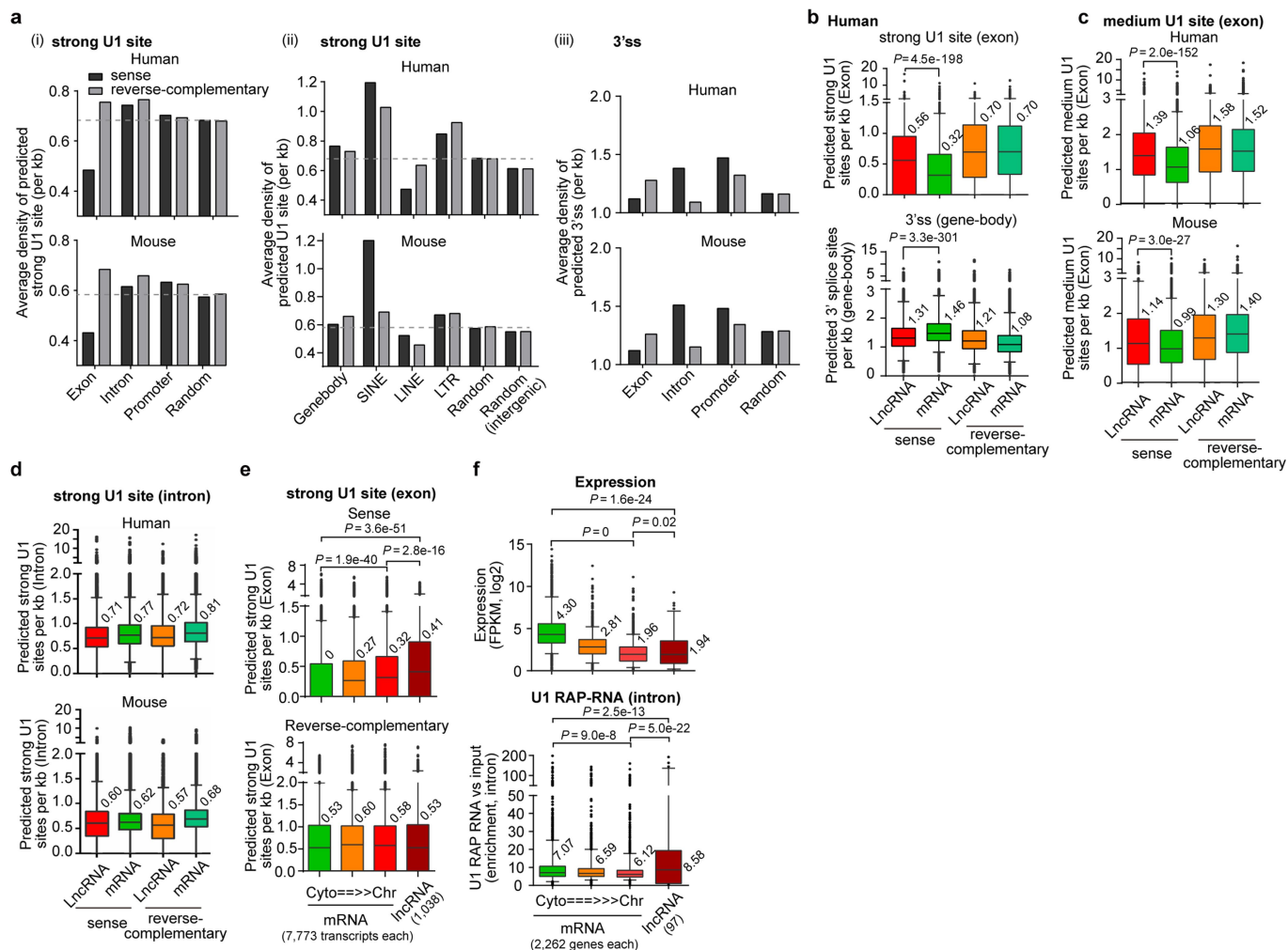


Extended Data Fig. 3 | See next page for caption.

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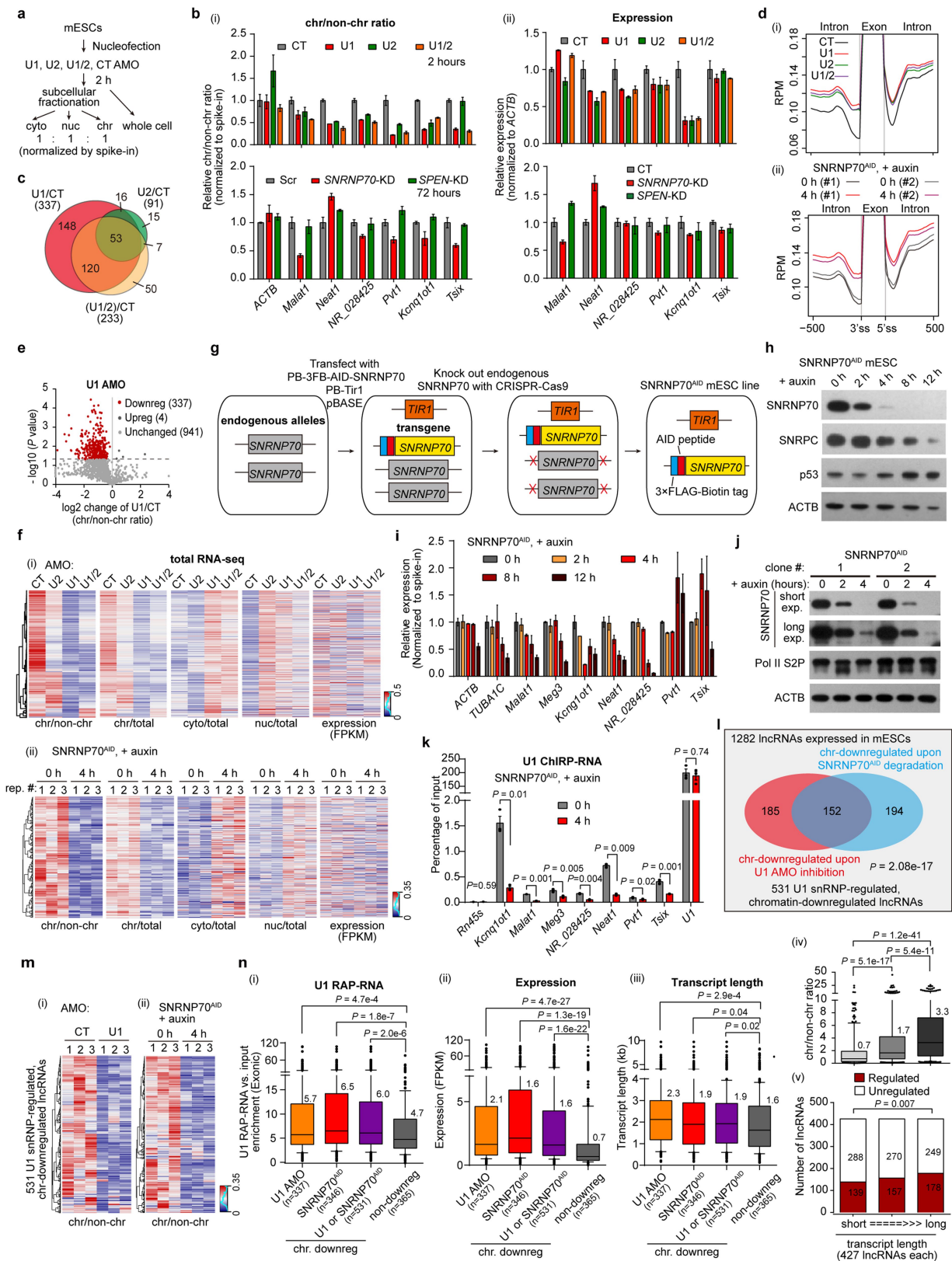
Extended Data Fig. 3 | Reporter assays reveal a key role of the U1 motif and U1 snRNP in RNA–chromatin association. **a**, Diagrams showing the *GFP* reporters that we constructed and analysed here. EGFP, enhanced GFP; EV, empty vector; PAS, bovine growth hormone (BGH) polyadenylation signal; 3' *Malat1*, 3' termination sequence of *Malat1*; U1 (28-nt), a short 28-nucleotide *NXF1*-enChr sequence that encompasses two U1-recognition motifs (one strong and one weak site from position 32 to 59 in *NXF1*-enChr). The *NXF1*-enChr *GFP* reporters (PAS and 3' *Malat1*) contain the 162-nucleotide *NXF1*-enChr sequences with either a wild-type U1-recognition site or a mutated site (mutU1, vertical black line). To rule out an indirect effect of RNA degradation on its chromatin localization, we replaced the PAS with the 3' termination sequence of *Malat1*. The 3' end of *Malat1* possesses a triple-helix structure, which resembles the viral expression and nuclear retention element (ENE) and stabilizes the *Malat1* transcripts^{21,22}. To assess the specificity of U1-mediated chromatin association, we also constructed *GFP* reporters carrying RepA alone and RepA with *NXF1*-enChr. We used two primer pairs (q1F/q1R and t1F/t1R; red arrows) to analyse the expression and potential splicing of the insert. **b**, The insertion of *NXF1*-enChr (U1) did not elicit splicing. The agarose gel picture shows PCR products amplified by primers t1F and t1R in mESCs expressing the *NXF1*-enChr (PAS) or *NXF1*-enChr-mutU1 (PAS) constructs. cDNAs were used as templates and the corresponding plasmids were used as control templates. $n = 2$ independent experiments. **c**, Subcellular fractionation and RT–qPCR analysis of *GFP* RNA in various reporters. Endogenous expression of *ACTB* and *Malat1* serves as internal controls. **d**, GFP fluorescence imaging of mESCs expressing the EV, *NXF1*-enChr (PAS) or *NXF1*-enChr-mutU1 (PAS) constructs shown in **a**. GFP fluorescence is much weaker in cells expressing the *NXF1*-enChr-PAS construct. $n = 4$ independent experiments. **e**, RT–qPCR analysis of chromatin/cytosol ratios (top) and relative expression (bottom) of *GFP* RNA in mESCs. The relative chromatin/cytosol ratio was normalized to a spike-in RNA that was prepared by in vitro transcription. Data shown as mean $\pm$ s.e.m.; $n = 2$ biological replicates. **f**, RT–qPCR analysis of the chromatin/non-chromatin ratio (i) and relative expression (ii) of *GFP* RNA in human HEK 293T and Hela cells. Similar results to those shown in **e** were observed in human cells, suggesting a conserved U1-based mechanism in humans and mice. **g**, Additive effect of

multiple enChrs in promoting RNA–chromatin association. The RepA-*NXF1*-enChr reporter exhibited significantly higher chromatin enrichment than RepA or *NXF1*-enChr alone. In **f**, **g**, data are shown as mean $\pm$ s.e.m.; P -values obtained by two-sided t -test with three biological replicates. **h**, Representative proteins identified by *NXF1*-enChr RNA pull-down assay. The mass-spectrometry scores of proteins identified by *NXF1*-enChr and *NXF1*-enChr-as RNA pull-down are shown, together with their fold enrichment in the *NXF1*-enChr sample relative to the antisense control (enChr-as). Sm proteins are general components of snRNPs that bind snRNAs. **i**, Western blot confirming the specific interaction between SNRNP70 and *NXF1*-enChr. Controls were the antisense sequence and the sequence with a deletion of the strong U1-recognition site (Δ U1). $n = 3$ independent experiments. **j**, RNAi in mESCs harbouring the *NXF1*-enChr (PAS) GFP reporter. Depletion, using short hairpin RNA (shRNA) knockdown (KD) of the three core components of U1 snRNP (*SNRNP70*, *SNRPA* and *SNRPC*), but not of *SPEN* or splicing regulators (*RBM6*, *U2AF1/2* and *SF3A2*), led to 2.5–4.5-fold increases in GFP signals analysed by FACS. **k**, Knockdown efficiency of *SNRNP70* using two shRNAs in mESCs. Top, RT–qPCR, mean $\pm$ s.e.m.; bottom, western blot. $n = 2$ independent experiments. **l**, **m**, RT–qPCR analysis of the chromatin/cytosol ratio (top) and relative expression (bottom panel) of *GFP* RNA in mESCs after knocking down *SNRNP70* (**l**) or *SPEN* (**m**) in mESCs expressing the *NXF1*-enChr-U1 reporter or RepA-*GFP* reporter. Means $\pm$ s.e.m. are shown. $n = 2$ biological replicates. Scr, scrambled control shRNA. **n**, Mutation analysis of the 5' and 3' splice sites using a *GFP* reporter carrying the intron-3 sequence of *ACTB*. **i**, The mutation scheme. **ii**, RT–qPCR analysis of relative chromatin/non-chromatin ratios (left) and expression (right) of *GFP* RNA in mESCs expressing different constructs. **iii**, PCR bands of spliced and unspliced transcripts. Splicing was abolished in the 3'ss and 5'+3'ss mutants. Splicing was detected in the 5'ss mutant reporter (red asterisk) owing to the presence of an alternative 5'ss downstream of the mutated site. **iv**, Results of sequencing PCR fragments of the 5'ss mutant reporter. Data are shown as mean $\pm$ s.e.m., and include three biological replicates. P -values obtained from two-sided t -test. See also Supplementary Notes 3 and 4. For gel source data, see Supplementary Fig. 1.



Extended Data Fig. 4 | Differential distributions of U1-recognition sites and 3' splice sites in mRNA and lncRNA genes. **a**, Average density of predicted strong U1-recognition sites in genic regions (i) and repeat elements (ii), and average density of predicted 3'ss in genic regions (iii), in humans (top panels) and mice (bottom panels). Random sequences, random intergenic sequences, and reverse-complementary sequences serve as controls for the background. LINE and SINE, long and short interspersed nuclear elements; LTR, long terminal repeat. **b**, Comparison of the density of predicted strong U1-recognition sites in the reverse-complementary strand of exons (top) and of 3'ss in the gene-body region (bottom) of lncRNAs (U1 site, $n = 4,731$; 3'ss, $n = 21,512$) and mRNA genes (U1 site, $n = 68,881$; 3'ss, $n = 139,458$) in humans. Only genes with transcript lengths (for U1 site) or genomic lengths (for 3'ss) larger than 1 kilobase were analysed. **c**, Densities of predicted medium-strength U1-recognition sites in exons of lncRNAs and mRNA genes in humans (top; lncRNA, $n = 4,731$; mRNA, $n = 68,881$) and mice (bottom; lncRNA, $n = 3,385$;

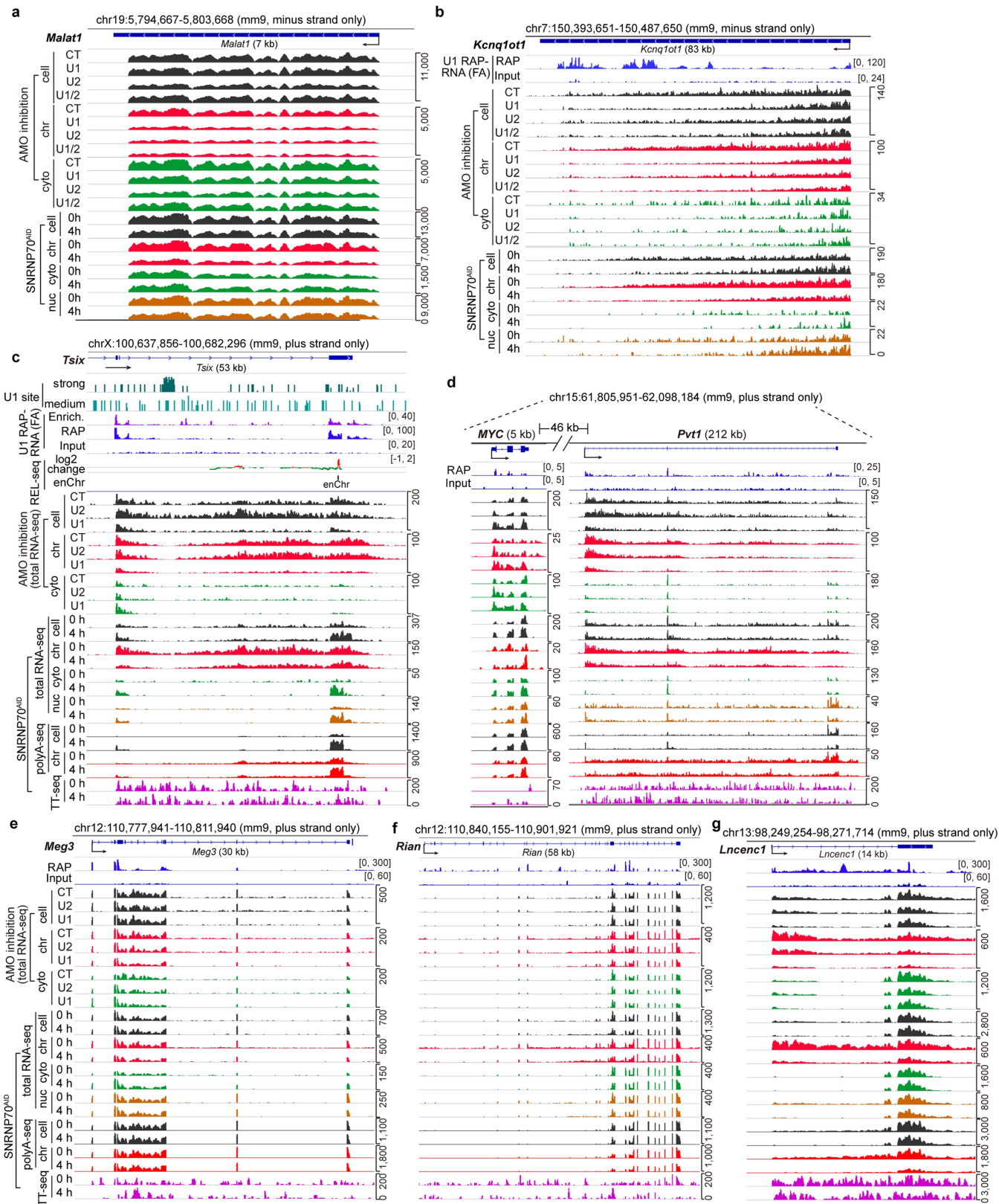
mRNA, $n = 47,298$). Transcripts with total exon lengths of more than 1,000 nucleotides were analysed. **d**, Densities of predicted strong U1-recognition sites in introns of lncRNA and mRNA genes in humans (top; lncRNA, $n = 15,885$; mRNA, $n = 130,431$) and mice (bottom; lncRNA, $n = 6,467$; mRNA, $n = 72,688$). **e**, Densities of predicted strong U1-recognition sites in the sense strand (top) and reverse-complementary strand (bottom) of mRNA with different levels of chromatin-binding activity ($n = 7,773$ each) and lncRNA ($n = 1,038$) transcripts (from low to high: green, low; orange, moderate; red, high; dark red, lncRNA). **f**, Comparison of expression (top) and fold enrichment of U1 RAP-RNA signals in introns (bottom) in mRNA genes ($n = 2,262$ each group) and lncRNAs ($n = 97$) with different levels of chromatin binding. LncRNAs that show detectable expression (FPKM values greater than 1) and no overlap with protein-coding genes were used. In **b-f**, P -values are based on two-sided t -tests. Box plots show 5th, 25th, 50th, 75th and 95th percentiles, with median values labelled by the box plots.



Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | U1 snRNP regulates the chromatin retention of lncRNAs. **a**, Scheme illustrating inhibition of U1 and/or U2 snRNA by AMO nucleofection followed by strand-specific RNA-seq analysis of total RNAs isolated from whole cells and from three subcellular fractions: cytoplasm (cyto), nucleoplasm (nuc) and chromatin (chr). CT, scramble control AMO. **b**, RT-qPCR of the chromatin/non-chromatin ratios (i) and relative expression (ii) of representative lncRNAs in mESCs after 2-h treatments with the control (CT), U1, U2 or U1/2 AMOs (upper panels), or after 72 h of knockdown with scramble (Scr), *SNRNP70* or *SPEN* shRNAs (bottom panels). (Non-chromatin refers to cytoplasm plus nucleoplasm.) Means $\pm$ s.e.m. are shown. $n = 2$ biological replicates. Consistent with inhibition of U1 snRNA, knockdown of *SNRNP70*—a core component of U1 snRNP—also led to decreased chromatin signals of individual lncRNAs. **c**, Venn diagram showing lncRNAs with decreased chromatin associations upon U1, U2 or U1/2 inhibition. Compared with U1 AMO (337), inhibition of U2 snRNA affected a smaller number (91) of lncRNAs, most of which (69) belong to U1-regulated lncRNAs. Inhibition of both U1/2 snRNA did not elicit a stronger effect than inhibition of U1 alone. Note that inhibition of both U1 and U2 used a lower concentration of U1 and U2 AMOs (50 μ M each) than the 75 μ M used to inhibit U1 alone. **d**, Metaplots of whole-cell RNA-seq reads in intron–exon junctions upon inhibition of U1 and/or U2 snRNA (i) or upon *SNRNP70*^{AID} degradation (ii). RPM, read count per million mapped reads. **e**, Volcano plot showing the fold change ($\log_2$) of the chromatin/non-chromatin ratio of lncRNAs ($n = 1,282$) upon inhibition of U1 snRNA (using AMOs, 2 h). P -values obtained by two-sided t -test with three biological replicates. **f**, Heat maps showing the ratio of chromatin/non-chromatin, the ratio of each fraction versus the normalized total RNA contents, and the expression level of 337 lncRNAs with decreased chromatin association upon U1 AMO inhibition (i); or 346 chromatin-downregulated lncRNAs in total RNA-seq upon *SNRNP70*^{AID} degradation (ii). Upon U1 AMO inhibition or *SNRNP70*^{AID} degradation, these lncRNAs show decreased chromatin association, while their relative abundance in cytoplasmic and nucleoplasm fractions even increased. **g**, Diagram showing the construction of the *SNRNP70*^{AID} mESC line, which expresses an AID- and FLAG-tagged *SNRNP70* (*SNRNP70*^{AID}) in a

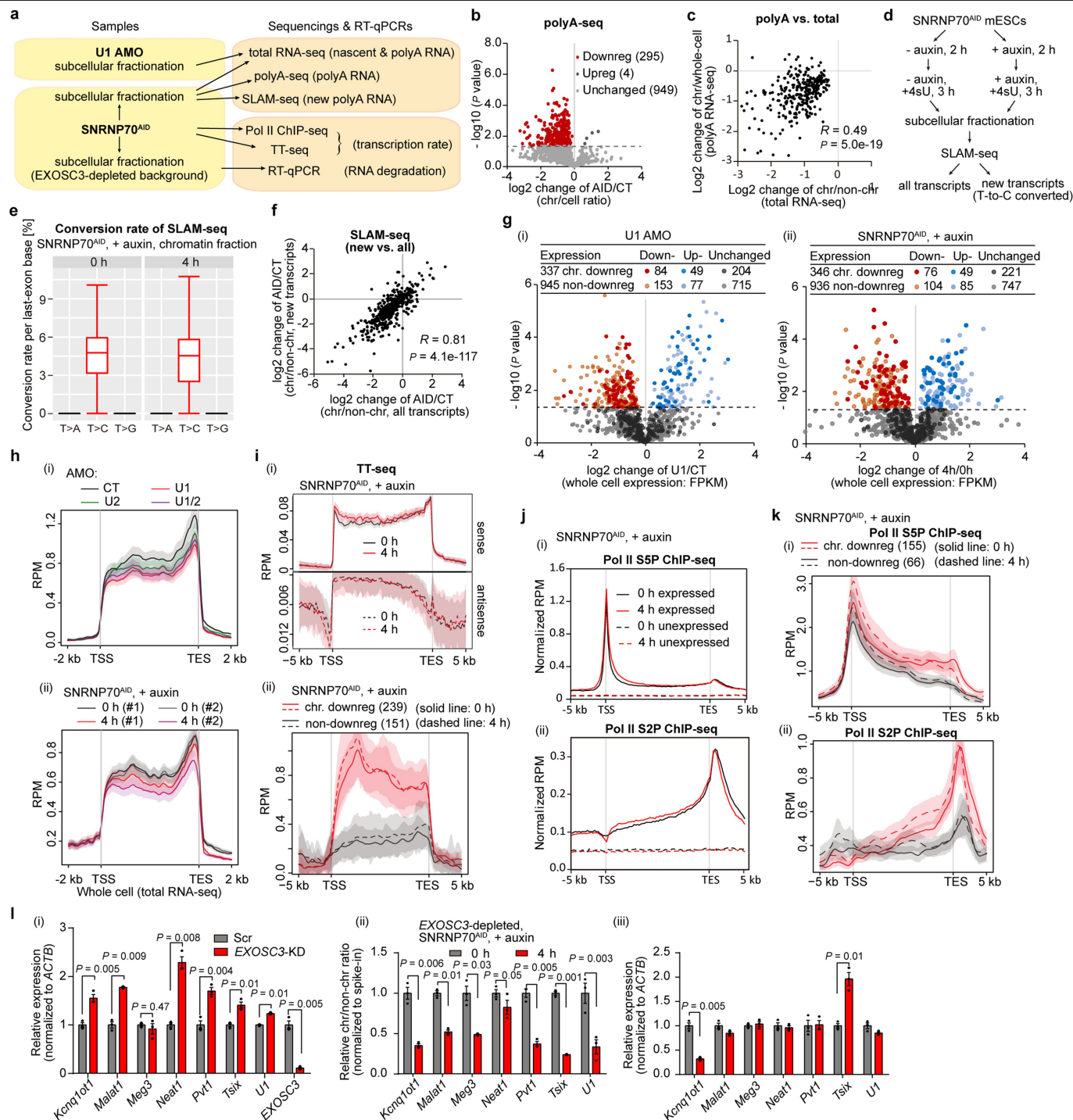
transgene with the two endogenous *SNRNP70* alleles inactivated. *TIRI*, DNA expressing E3 ligase for the AID system. **h**, **i**, Time-course expression analysis by western blot (**h**) and by RT-qPCR (**i**) in *SNRNP70*^{AID} mESCs upon addition of auxin for 0–12 h. $n = 2$ independent experiments. For panel **i**, means $\pm$ s.e.m. are shown. RNA expression was normalized by a *lacZ* spike-in that was added into the same numbers of cells. At 4 h of auxin treatment, the expression of p53 protein and the RNA transcripts analysed exhibited modest changes compared to the changes after 8–12 h. **j**, Auxin-induced rapid degradation of *SNRNP70*^{AID} did not affect the phosphorylation of Pol II at Ser 2 (S2P). ACTB serves as a loading control. $n = 3$ independent experiments. **k**, RT-qPCR analysis showing enrichments of candidate lncRNAs captured by U1 ChIRP before (0 h) and after (4 h) *SNRNP70*^{AID} depletion. U1 binding to its target candidate lncRNAs was severely impaired in auxin-treated mESCs (4 h), indicating that the function of U1 snRNA requires an intact snRNP complex. Data shown as mean $\pm$ s.e.m., from three biological replicates. P -values obtained by two-sided t -test. **l**, The overlap of a total of 531 U1-snRNP-regulated, chromatin-downregulated lncRNAs from U1 AMO inhibition (red) or *SNRNP70*^{AID} degradation (blue). P -values obtained by exact hypergeometric probability. **m**, Heat map showing changes of the chromatin/non-chromatin ratio upon U1 AMO inhibition (i) or degradation of *SNRNP70*^{AID} (ii) for the set of 531 U1-snRNP-regulated, chromatin-downregulated lncRNAs shown in **l**. The patterns of chromatin/non-chromatin ratio changes are highly similar for both treatments. **n**, Analysis of U1 RAP–RNA signals (i), expression levels (FPKM) (ii) and transcript lengths (iii–v) in the various sets of lncRNAs shown in **l**. Box plots show 5th, 25th, 50th, 75th and 95th percentiles, with median values labelled by the box plots and sample sizes ($n \geq 337$) labelled on the x axis. We further divided mESC-expressed lncRNAs into three groups on the basis of their transcript length (from shortest to longest). iv, v, The chromatin/non-chromatin ratio of total RNA-seq (iv) and the numbers of U1-snRNP-regulated and unregulated lncRNAs (v) in each group ($n = 427$). Longer lncRNAs appear to exhibit stronger chromatin retention and to be preferentially affected upon *SNRNP70*^{AID} degradation. P -values obtained by two-sided Mann–Whitney test for i–iv, one-sided Fisher’s exact test for v. For gel source data, see Supplementary Fig. 1.



Extended Data Fig. 6 | See next page for caption.

Extended Data Fig. 6 | Sequencing tracks of representative lncRNA and mRNA genes. **a**, *Malat1*. **b**, *Kcnq1ot1*. **c**, *Tsix*. **d**, *Pvt1* and its upstream protein-coding gene, *MYC*. **e**, *Meg3*. **f**, *Rian*. **g**, *Lncenc1*. In the *Tsix* locus (**c**), the top three sets of tracks show the predicted U1-recognition sites (strong or medium, indicated by blue vertical lines), U1RAP-RNA-seq with formaldehyde (FA) crosslinking (including enrichment ratio, reads signals of U1 RAP and the input control), and the REL-seq result. The lower two sets of tracks show total RNA-seq after AMO treatments, and total and polyA RNA-seq and TT-seq of SNRNP70^{ΔID} mESCs at 0 h or 4 h of auxin treatment. Total RNA-seq analysis revealed decreased chromatin levels of *Tsix* transcripts after inhibition of U1 snRNA or degradation of SNRNP70^{ΔID}. Intriguingly, polyA-seq showed a more dramatic increase of polyadenylated *Tsix* transcripts in the whole-cell sample compared with that in the chromatin fraction upon SNRNP70^{ΔID} degradation. Coincidentally, an U1-associated enChr (black vertical line) was identified by REL-seq at roughly 2.2 kb upstream of the annotated transcription end site

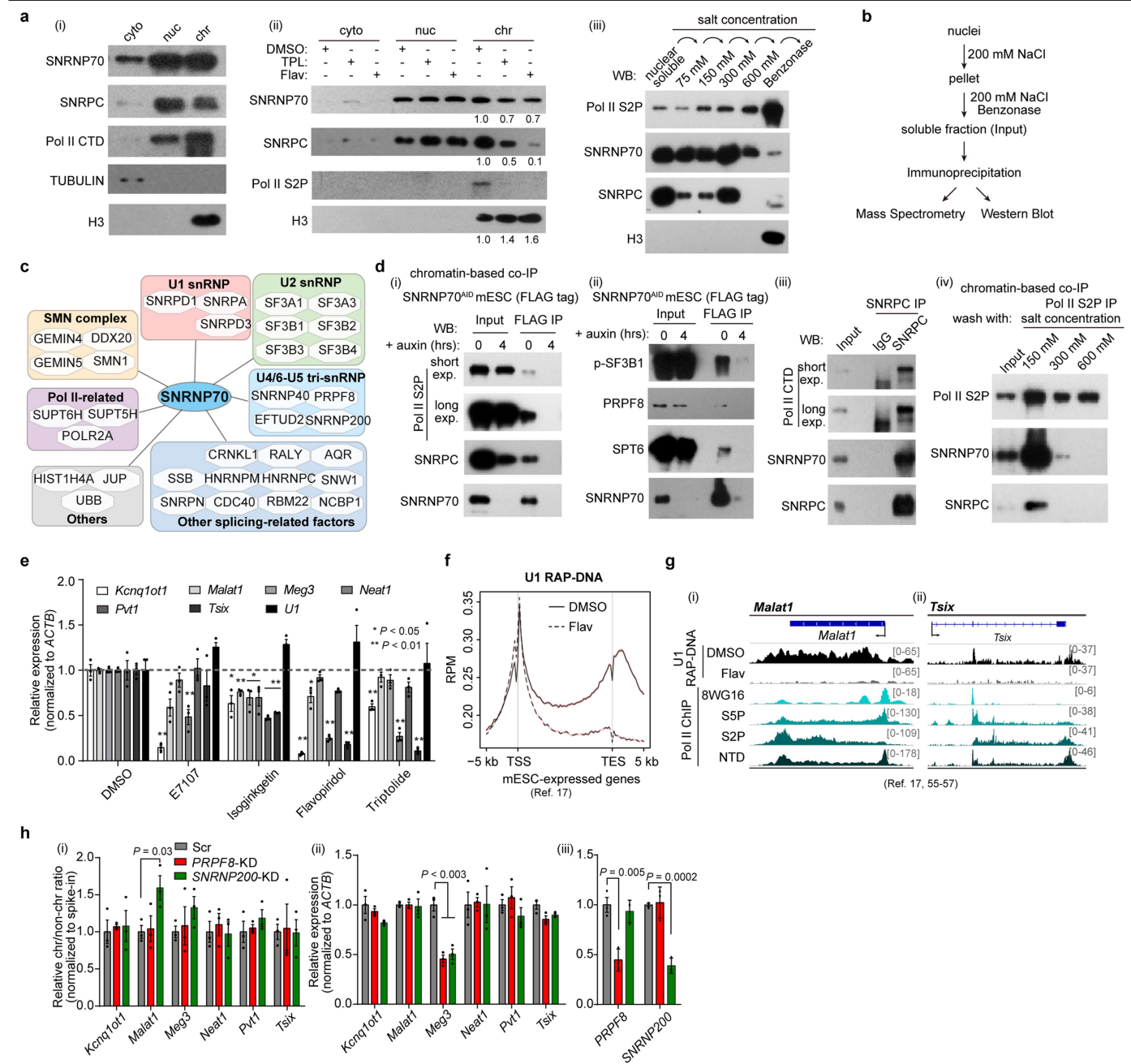
(TES) of *Tsix*. In addition, strong binding of U1 snRNA was detected extensively across the whole exon at the 3' end of *Tsix*. Thus, U1 snRNP may inhibit the PAS of *Tsix* to promote its degradation and chromatin retention. Inhibition of U1 snRNP by degradation of SNRNP70^{ΔID} thus appears to enhance polyadenylation, stability and nuclear export of *Tsix*, leading to the observed increases of polyA RNA in both whole-cell and chromatin fractions. Nevertheless, the chromatin/whole-cell ratio of polyA *Tsix* still decreases after SNRNP70^{ΔID} degradation. Two well spliced lncRNAs *Meg3* (**e**) and *Rian* (**f**) show few intronic signals in both total and polyA RNA-seq. In **d-g**, U1RAP-RNA-seq (FA), total RNA-seq after AMO treatments, and total and polyA RNA-seq and TT-seq tracks of SNRNP70^{ΔID} mESCs at 0 h or 4 h auxin treatment are shown. Some very large lncRNAs—such as *Kcnq1ot1* (**b**, 83.4 kb in the genome sequence), *Tsix* (**c**, 53.4 kb) and *Pvt1* (**d**, 213 kb)—show decreased RNA signals in the downstream gene body upon U1 AMO treatment; however, this effect was less obvious in SNRNP70^{ΔID} mESCs.



Extended Data Fig. 7 | See next page for caption.

Extended Data Fig. 7 | Analysis of the direct causality of U1 snRNP in regulating lncRNA–chromatin retention. **a**, Summary of experiments that we carried out to systematically investigate the effects of acute inhibition of U1 snRNP on lncRNA–chromatin associations, transcription dynamics, and RNA processing and decay. **b**, Volcano plot of polyA RNA-seq showing the fold change ($\log_2$) in the chromatin/whole-cell ratio of lncRNAs upon SNRNP70^Δ degradation. Red dots and deep grey dots indicate lncRNAs that show a significant decrease or increase, respectively, in their chromatin/cell ratio by comparing SNRNP70^Δ (4-h auxin treatment) versus control (0 h) ($P < 0.05$; two-sided t -test with three biological replicates). **c**, Correlation plot of polyA and total RNA-seq analysis of SNRNP70^Δ mESCs. The set of chromatin-downregulated lncRNAs shows significantly correlated changes in chromatin localization upon degradation of SNRNP70^Δ ($n = 346$). **d**, SLAM-seq analysis of chromatin and non-chromatin (cytoplasm and nucleoplasm) fractions in SNRNP70^Δ mESCs. SNRNP70^Δ mESCs were treated with or without auxin for 2 h and then labelled with 4sU for 3 h. After chemical conversion of the incorporated 4sU nucleotides to cytidine, RNA from various subcellular fractions was isolated for 3'-end polyA-seq library construction. **e**, Box plots showing the conversion rate detected by SLAM-seq of chromatin fractions in the last exon of genes with detectable new transcript ($n = 24,097$) before (0 h) and after (4 h) SNRNP70^Δ degradation. Box plots show 5th, 25th, 50th, 75th and 95th percentiles. **f**, Pearson correlation analysis of the change in chromatin/non-chromatin ratio for new versus all transcripts of lncRNAs with detectable new transcripts ($n = 492$) identified by SLAM-seq. **g**, Volcano plots showing expression changes of mESC-expressed lncRNAs ($n = 1,282$) after treatment with U1AMO (i) or degradation of SNRNP70^Δ (ii). Chromatin-downregulated or non-downregulated lncRNAs were further classified into 'downregulated (down-)', 'upregulated (up-)' or 'unchanged' according to their expression changes in whole-cell samples. lncRNAs with reduced chromatin association upon inhibition of U1 snRNA or SNRNP70^Δ do not show greater downregulated expression by comparison with all lncRNAs. Only a small proportion of them (84 of 337 U1-regulated and 76 of 346 SNRNP70-regulated) show decreased transcript levels. P -values obtained by two-sided t -test; $n = 3$ biological replicates. **h**, Metagene analysis of whole-cell RNA-seq reads for the

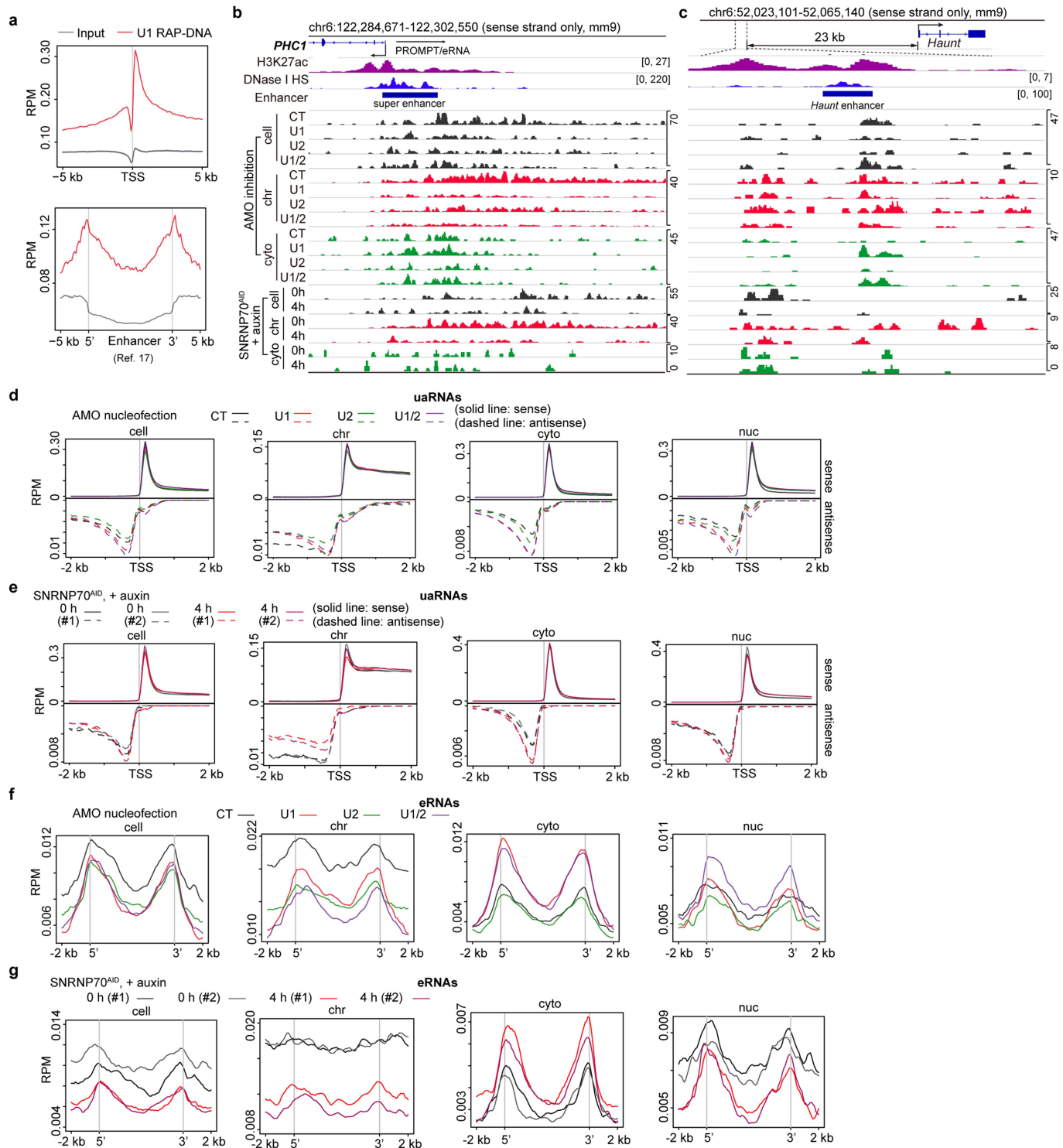
set of U1-snRNP-regulated, chromatin-downregulated lncRNAs in mESCs. Only lncRNAs that do not overlap with any protein-coding gene on the same strand were analysed ($n = 239$). Similar read-distribution patterns were observed in control cells and in cells treated with U1, U2 or U1/U2 AMOs (i) or in cells subjected to auxin-induced degradation of SNRNP70^Δ for 0 h or 4 h (ii). Thus, rapid inhibition of U1 snRNP did not cause global transcription termination, although we did observe decreased downstream RNA signals in a few very long lncRNAs, such as *Kcnq1ot1* (83 kb), in agreement with the proposed role of U1 telescripting in protecting the transcription integrity of very large transcripts²⁵. In addition, we conjecture that the slight decreases in total transcript levels are likely to be post-transcriptionally mediated by RNA degradation instead of an effect of U1 inhibition on nascent transcription (see panels **i–k** below). **i**, Metagene analysis of TT-seq signals in all mESC-expressed genes ($n = 10,675$) (i) and chromatin-downregulated ($n = 239$) and non-downregulated lncRNAs ($n = 151$) (ii) upon SNRNP70^Δ degradation. Only lncRNAs that do not overlap with any protein-coding gene on the same strand were analysed. **j, k**, Metagene analysis of ChIP-seq signals of Pol II S5P ('paused' Pol II; i) and S2P ('elongating' Pol II; ii) across the gene body and upstream/downstream 5-kb region of all mESC-expressed genes ($n = 10,675$) and unexpressed genes ($n = 7,933$) (**j**) or chromatin-downregulated ($n = 155$) and non-downregulated ($n = 66$) lncRNAs upon SNRNP70^Δ degradation (**k**). Only lncRNAs that do not overlap with any protein-coding gene on either strand were analysed. For **h–k**, shadings represent 95% confidence intervals for the average enrichment. **l**, RT-qPCR analysis of the knockdown efficiency and lncRNA expression level change (i), chromatin/non-chromatin ratio (ii) and relative expression (iii) in SNRNP70^Δ mESCs depleted of *EXOSC3* by RNA inhibition. Knockdown was analysed at 72 h after shRNA viral infection. The observation of increased expression for most lncRNAs analysed is consistent with a role of *EXOSC3* in mediating RNA degradation (i). Knockdown of *EXOSC3* blocked RNA degradation for most lncRNAs analysed (iii), but failed to rescue their decreased chromatin associations induced by auxin (ii). Thus, the effect of U1 snRNP in promoting lncRNA–chromatin binding is not caused by increased RNA degradation. Means $\pm$ s.e.m. are shown; P -values obtained by two-sided t -test for three biological replicates.



Extended Data Fig. 8 | See next page for caption.

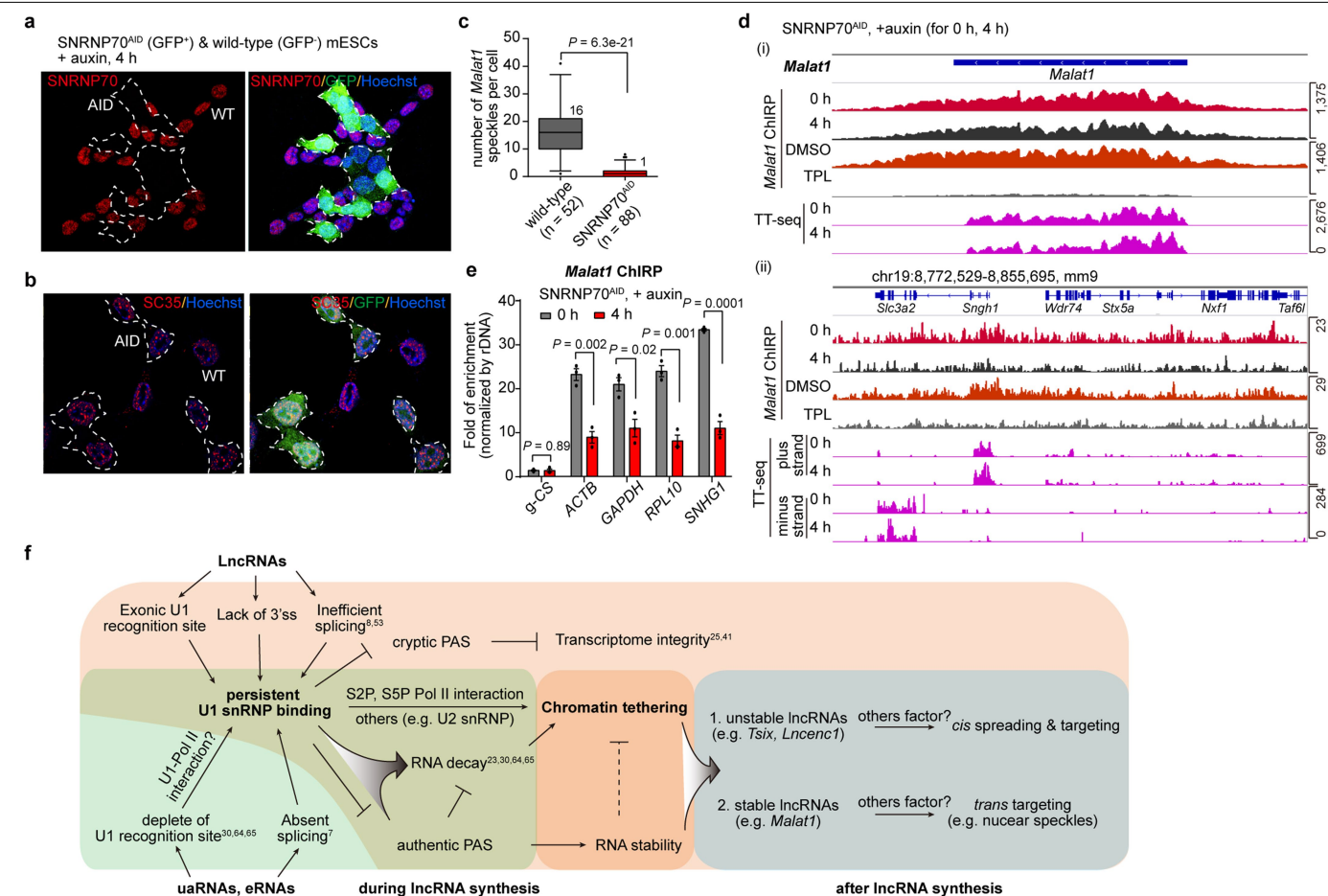
Extended Data Fig. 8 | U1 snRNP regulates lncRNA–chromatin association through its interaction with transcriptionally engaged Pol II. **a**, Western blots of SNRNP70, SNRPC, Pol II and marker proteins with no treatment (i), or treated with DMSO, triptolide (TPL), or flavopiridol (Flav) for 1 h (ii), or in nuclear fractions sequentially extracted with increasing concentrations of salt (NaCl) (iii). For i, iii, $n = 3$ independent experiments, for ii, $n = 2$ independent experiments. **b**, Scheme for co-immunoprecipitation of the chromatin fraction. Benzonase was used to digest all DNA and RNA and to elute proteins from chromatin. **c**, Proteins captured by SNRNP70 coIP-SILAC. Native chromatin extracts that were released by benzonase were subjected to anti-FLAG co-immunoprecipitation of SNRNP70^{AID} protein (FLAG-tagged) coupled with stable-isotope labelling by amino acids in cell culture, followed by mass spectrometry. SNRNP70 purification captured the U1 and U2 snRNPs, as well as several components of the U4/6-U5 snRNPs and other splicing factors. Notably, the SNRNP70 interactome also identified proteins involved in transcription regulation, such as POLR2A, SPT5 and SPT6. **d**, Chromatin-based co-immunoprecipitation (in the presence of benzonase) and western blot analysis confirming the interactions between U1 snRNP and the proteins identified in **c**. Panels i, ii show co-immunoprecipitation of FLAG-tagged SNRNP70^{AID} protein. SNRNP70^{AID} mESCs treated with auxin for 4 h serve as the negative control. Panel iii shows that co-immunoprecipitation of endogenous SNRPC captured

the total Pol II. Panel iv shows that co-immunoprecipitation of Pol II S2P captured SNRNP70 under a physiological salt condition (150 mM) but not in high-salt conditions, suggesting dynamic associations between U1 snRNP and engaged Pol II. $n = 3$ independent experiments. **e**, RT-qPCR analysis of the expression of representative lncRNAs and U1 snRNA upon treatment with DMSO (control), 1 μ M with E7107 (inhibitor of U2 snRNP), 1 μ M of flavopiridol (transcription inhibitor) or 1 μ M of triptolide (transcription inhibitor) for 1 h, or with 100 μ M of isoginkgetin (inhibitor of U4/6-U5 snRNP) for 2 h. Means $\pm$ s.e.m. are shown; P -values obtained by two-sided t -test with three biological replicates. **f**, Metagene analysis of U1 RAP–DNA-seq signals for all *Ensembl* genes in mESCs treated with DMSO or with flavopiridol (Flav) for 1 h. Flavopiridol treatment led to reduced U1 RAP–DNA signals downstream of the TSS across the gene body. Red shading represents 95% confidence intervals for the average enrichment. **g**, Genome-browser views of U1 RAP–DNA-seq and Pol II ChIP-seq at the *Malat1* and *Tsix* loci. SWG16, hypophosphorylated Pol II; NTD, N-terminal domain of Pol II, representing the total Pol II. **h**, Relative chromatin/non-chromatin ratio (i), expression-level change of representative lncRNAs (ii) and knockdown efficiency of targeted genes (iii) after mESCs were treated with *PRPF8* or *SNRNP200* shRNAs for 72 h. Means $\pm$ s.e.m. are shown; P -values obtained by two-sided t -test with three biological replicates. For gel source data, see Supplementary Fig. 1.



Extended Data Fig. 9 | Inhibition of U1 and U2 snRNPs downregulates the chromatin association of uaRNAs and eRNAs. **a**, Metaplots of U1 RAP-DNA-seq, showing enrichment of U1 snRNA in the chromatin proximity of regulatory DNA sequences. The top panel shows a ± 5 -kb window flanking TSSs of *Ensembl* genes that do not overlap with any other gene within 2 kb ($n = 18,972$), and the bottom panel shows a ± 5 -kb window flanking enhancers that do not overlap with a gene within 2 kb ($n = 3,767$). **b**, **c**, Sequencing tracks showing chromatin and cytoplasmic RNA-seq signals of uaRNAs/eRNAs in the *PHC1* promoter (**b**) and *Haunt* enhancer (**c**). **d**, **e**, Metaplots of RNA-seq reads of uaRNAs from whole cells and subcellular fractions in AMO-treated samples (**d**) and

SNRNP70^{AID}-degraded samples (**e**) in a ± 2 -kb window flanking the TSSs of *Ensembl* genes that do not overlap with any other gene within 2 kb ($n = 18,972$). The uaRNAs show upregulated overall expression and more dramatic increases in the cytoplasmic and nucleoplasmic fractions after U1 or U1/2 inhibition, while there are comparable (U1 AMO) or slightly decreased (U1/2 AMO) uaRNA signals in the chromatin fraction at the TSS-to-1-kb upstream region. **f**, **g**, Metaplots of RNA-seq reads of eRNAs from whole cells and subcellular fractions in AMO-treated samples (**f**) and SNRNP70^{AID}-degraded samples (**g**) in a ± 2 -kb window flanking enhancers that do not overlap with any gene within 2 kb ($n = 3,767$).



Extended Data Fig. 10 | U1 snRNP tethers and mobilizes lncRNAs to chromatin. **a, b**, Immunofluorescence analysis of SNRNP70 (**a**) and SC35 (**b**) in auxin-treated SNRNP70^{AID} and wild-type mESCs. SNRNP70^{AID} ESCs marked by stably integrated GFP transgenes (GFP⁺, highlighted with dashed white lines) and GFP-negative wild-type mESCs (GFP⁻) were mixed and treated with auxin for 4 h. $n = 3$ independent experiments. **c**, Quantification of the numbers of *Malat1* speckles (equivalent diameter greater than 0.5 μm) identified by RNA FISH (Fig. 3c). Box plots show 5th, 25th, 50th, 75th and 95th percentiles, with median values labelled by the plots and sample sizes (wild type, $n = 52$; SNRNP70^{AID}, $n = 88$) labelled on the x axis. P -values obtained by two-sided Mann-Whitney test. **d**, Sequencing tracks of *Malat1* ChIP-DNA-seq and TT-seq in *Malat1* (i) and representative loci that are targeted by *Malat1* (ii). In both panels, the top set of tracks show *Malat1* ChIP-seq upon SNRNP70^{AID} degradation (at 0 h and 4 h) or upon treatment with DMSO control or triptolide (TPL). The bottom set of tracks show TT-seq upon SNRNP70^{AID} degradation (at 0 h and 4 h). In panel ii, TT-seq signals on both plus and minus strands are shown. We used the mm9 mouse genome assembly. **e**, qPCR analysis of *Malat1* ChIP-DNA of SNRNP70^{AID} mESCs before (0 h) and after (4 h) treatment with auxin. Data are shown as mean $\pm$ s.e.m., for three biological replicates. P -values obtained by two-sided t -test. **f**, Mechanistic representation of U1 snRNP and its interplay with Pol II and PASs in regulating the tethering and mobilization of noncoding RNA on chromatin. Notably, lncRNAs, uaRNAs and eRNAs share many features, including chromatin association, inefficient or absent splicing and polyadenylation, low-level expression and short half-lives^{7,8,42,53,63}. lncRNAs in general are enriched with 5' U1-recognition sites but depleted of 3' splice sites. For uaRNAs and eRNAs, U1 binding on chromatin is enriched at enhancer DNA sequences and TSSs (the 5' end of uaRNAs), even though U1-recognition

sites are depleted in uaRNA DNA sequences^{17,30,64,65}. U1 snRNP may bind uaRNAs and eRNAs through co-transcriptional U1–Pol II interactions. Splicing releases the U1 snRNP from pre-mRNAs¹⁸. However, lncRNAs, uaRNA and eRNAs remain associated with U1 snRNPs because of inefficient or absent splicing (the lack of 3' ss could be one reason)^{8,53}. Through its interaction with transcriptionally engaged Pol II, U1 snRNP is tethered to chromatin and subsequently retains its associated lncRNAs and ncRNAs on chromatin. Meanwhile, the inhibitory function of U1 snRNP on polyadenylation promotes transcription elongation at cryptic PASs and RNA decay at authentic PASs^{23,25,30,41,65}. Rapid RNA turnover renders these transcripts less likely to leave the chromatin, contributing in part to their enrichment on chromatin and lack of nuclear export. Although the properties of chromatin binding and instability appear to be intrinsically coupled for lncRNAs and chromatin-bound unstable ncRNAs, U1 snRNP and the RNA-degradation machinery appear to play independent yet synergistic roles in facilitating RNA–chromatin association. Coupled chromatin association and instability of many lncRNA transcripts may contribute to the observed *cis*-targeting and regulatory functions in their chromatin neighbourhoods. Most short-lived lncRNA transcripts spread locally within their neighbourhoods, while a few stable and abundant lncRNAs, such as *Malat1*, exist long enough to be trans-targeted to other genomic sites. For stable lncRNAs, persistent binding with U1 snRNP, and perhaps engaged Pol II, may drive lncRNA mobilization to distinct nuclear compartments (such as nuclear speckles) or to thousands of *trans* genomic sites (in the case of *Malat1*). Possibly, these highly expressed lncRNAs have developed evolutionarily to take advantage of the U1-tethering mechanism to achieve *trans* functions. In addition to U1 snRNP, U2 snRNP (but not the splicing reaction), and perhaps other factors, contributes to this process.

Reporting Summary

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Statistical parameters

When statistical analyses are reported, confirm that the following items are present in the relevant location (e.g. figure legend, table legend, main text, or Methods section).

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ An indication of whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☒ ☐ A description of all covariates tested
- ☒ ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistics including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated
- ☐ ☒ Clearly defined error bars
State explicitly what error bars represent (e.g. SD, SE, CI)

Our web collection on [statistics for biologists](#) may be useful.

Software and code

Policy information about [availability of computer code](#)

Data collection

Bio-Rad CFX384 Real Time System was used for RT-qPCR data collection. Illumina HiSeq 2500 and HiSeq X TEN were used for sequencing data collection. nikon A1RMP microscopy and related softwares were used for collecting and analyzing FISH and IF data.

Data analysis

For RNA-seq, REL-seq, TT-seq, and ChIP-seq data analysis, Bowtie v2.2.3, Bowtie v1.0.0, Tophat v2.0.10, Cufflink v2.1.1, BEDTools v2.17.0, macs v1.4.2 and Novoalign v3.02.05 were used. Integrative Genomics Viewer v2.3.97 were used for signal visualization of sequencing data. R studio v3.4.3, Excel 2016, and GraphPad Prism v6.0 were used for statistical analysis. FlowJo v7.6.1 was used for FACS analysis. Image J v1.8.0 was used for quantification of Western blot. "RNAfold v2.4.13 web server" (<http://rna.tbi.univie.ac.at/cgi-bin/RNAWebSuite/RNAfold.cgi>) was used for RNA secondary structure predication.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers upon request. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

The main data supporting the findings of this study are available within the article and Supplementary Information files. Sequencing data have been deposited in the GEO database under the accession number GEO: GSE107131 and GSE134287, and will be available after the manuscript is formally accepted. All other data supporting the findings of this study are available from the corresponding author upon reasonable request.

Field-specific reporting

Please select the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/authors/policies/ReportingSummary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical calculations were performed to estimate sample size. For REL-seq analysis, we included eight libraries and two representative cell lines (mouse embryonic stem cells and human HEK 293T), which are sufficient for screening technologies. See methods and supplementary note 1. For RNA-seq, we included three independent biological replicates, which are sufficient for sequencing analysis. Sample size determination in RT-qPCR and other graphs were described as previous published study and experimental knowledge.
Data exclusions	No data were excluded.
Replication	All experiments were repeated (biological repeats) at least twice. All attempts of replications were successful.
Randomization	There is not any bias when harvesting samples or collecting data.
Blinding	Experiments in this study were not done blinded. The samples harvest process made it impossible to be blinded. Data were collected in parallel to minimize bias.

Reporting for specific materials, systems and methods

Materials & experimental systems

n/a	Involved in the study
☐	☒ Unique biological materials
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☒	☐ Human research participants

Methods

n/a	Involved in the study
☐	☒ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Unique biological materials

Policy information about [availability of materials](#)

Obtaining unique materials All unique materials generated and described in this study will be provided upon reasonable request to the corresponding authors.

Antibodies

Antibodies used

Antibodies used for IP and western blot: SNRNP70 (Santa Cruz, sc-390988, E-4, lot: D0915, 1:500); SNRPC (Abcam, ab192028, EPR16034, lot: GR180605-1, 1:5000); Histone H3 (Easybio, BE2013-100 lot:80920815, 1:10000); TUBULIN (CWBIO, CW0098M, 1:2000); ACTB (Abclonal, AC004, lot: 01270/30337, 1:5000); RNAPII CTD (Abcam, ab52202, 1:2000); Hypophosphorylated RNAPII (Covance, MMS-126R, 8WG16, 1:2000 for western blot 1:200 for IP); RNAPII Ser5P (CST, 13523, D9N5I, lot: 1, 1:5000 for western blot; 1:300 for IP and ChIP); RNAPII Ser2P for IP and western blot (CST, 13499, E1Z3G lot: 1, 1:5000 for western blot; 1:300 for IP and ChIP), RNAPII Ser2P for western blot (Covance, MMS-129R, H5, lot: 14943202, 1:1000); p53 (Abclonal, A5761, 1:1000); SNRNP200 (Abclonal, A6063, lot: 0102340101, 1:1000); PRPF8 (Sangon, D163565, lot: D163565-0025, 1:1000); p-SF3B1 (Abclonal, AP0844, lot: 2158550301, 1:2000); Spt6 (CST, 15616S, D6J9H, lot: 1, 1:2000); SC35 (abcam, ab11826, 1:500 for IF).

Validation

Anti-SNRNP70 (Santa Cruz, sc-390988, E-4, lot: D0915), this antibody has been claimed to react with mouse SNRNP70 by the manufacturer. Validation data for the antibodies used can be found as follows: <https://www.scbt.com/p/u1-snrnp-70-antibody-e-4>.

Anti-SNRPC (Abcam, ab192028, EPR16034), this antibody has been claimed to react with mouse .. by the manufacturer. Validation data for the antibodies used can be found as follows: <https://www.abcam.com/u1-c-antibody-epr16034-ab192028.html>.

Anti-Histone H3 (Easybio, BE3021-100), this antibody has been claimed to react with mouse Histone H3 by the manufacturer (<http://www.bioeasytech.com/home/product/article/id/11.html>) and also confirmed by our data.

Anti-TUBULIN (CWBIO, CW0098M), this antibody has been claimed to react with mouse TUBULIN by the manufacturer. Validation data for the antibodies used can be found as follows: <https://www.cwbio.com/uploads/websitespdf/98be23fc-9451-4a27-93f6-02375421a1e5.pdf>

Anti-ACTB (Abclonal, AC004), this antibody has been claimed to react with mouse ACTB by the manufacturer. Validation data for the antibodies used can be found as follows: <https://abclonal.com.cn/catalog/AC004>.

Anti-RNAPII CTD (Abcam, ab52202), this antibody has been claimed to react with the CTD of mouse Polymerase II by the manufacturer. Validation data for the antibodies used can be found as follows: <https://www.abcam.com/rna-polymerase-ii-ctd-repeat-ysptps-antibody-ab52202.html>. This antibody has been used by Li et al, 2015 to detect Pol II by western blot, and also confirmed by our data.

Anti-Hypophosphorylated RNAPII (Covance, MMS-126R, 8WG16), this antibody has been claimed to react with mouse Polymerase II by the manufacturer. This antibody has been used for ChIA-PET by Bertolini et al, 2019, and also confirmed by our data.

Anti-RNAPII Ser5P (CST, 13523, D9N5I), this antibody has been claimed to react with mouse Ser5 phosphorylated Pol II by the manufacturer. Validation data for the antibodies used can be found as follows: <https://www.cellsignal.com/products/primary-antibodies/phospho-rpb1-ctd-ser5-d9n5i-rabbit-mab/13523>. This antibody has been used by Jiang et al, 2018 for ChIP assay, and also confirmed by our data.

Anti-RNAPII Ser2P (CST, 13499, E1Z3G), this antibody has been claimed to react with mouse Ser2 phosphorylated Pol II by the manufacturer. Validation data for the antibodies used can be found as follows: <https://www.cellsignal.com/products/primary-antibodies/phospho-rpb1-ctd-ser2-e1z3g-rabbit-mab/13499>.

Anti-RNAPII Ser2P (Covance, MMS-129R, H5), this antibody has been claimed to react with mouse Ser2 phosphorylated Pol II by the manufacturer. This antibody has been used for ChIP by Espinosa et al, 2003, and also confirmed by our data.

Anti-p53 (Abclonal, A5761), this antibody has been claimed to react with mouse p53 by the manufacturer. Validation data for the antibodies used can be found as follows: <https://abclonal.com.cn/catalog/A5761>.

Anti-SNRNP200 (Abclonal, A6063), this antibody has been claimed to react with mouse SNRNP200 by the manufacturer. Validation data for the antibodies used can be found as follows: <https://abclonal.com.cn/catalog/A6063>.

Anti-PRPF8 (Sangon, D163565), this antibody has been claimed to react with mouse PRPF8 by the manufacturer. Validation data for the antibodies used can be found as follows: <https://www.sangon.com/productDetail?productInfo.code=D163565>.

Anti-p-SF3B1 (Abclonal, AP0844), this antibody has been claimed to react with mouse phosphorylated SF3B1 by the manufacturer. Validation data for the antibodies used can be found as follows: <https://abclonal.com.cn/catalog/AP0844>.

Anti-SPT6 (CST, 15616S, D6J9H), this antibody has been claimed to react with mouse SPT6 by the manufacturer. Validation data for the antibodies used can be found as follows: <https://www.cellsignal.com/products/primary-antibodies/spt6-d6j9h-rabbit-mab/15616>.

Anti-SC35 (abcam, ab11826), this antibody has been claimed to react with mouse SC-35 by the manufacturer. Validation data for the antibodies used can be found as follows: <https://www.abcam.com/sc35-antibody-sc-35-nuclear-speckle-marker-ab11826.html>. This antibody has been used for IF by Gavrilov et al, 2013, and also confirmed by our data.

Eukaryotic cell lines

Policy information about cell lines

Cell line source(s)

HEK293T cells were obtained from ATCC (CRL-3216), Hela cells were obtained from ATCC (CCL-2), and mouse ESCs (E14T, 46C) was a gift from Austin Smith's lab.

Authentication

The cell lines have been used in the lab for over 3 years, so authentications were not performed. The SNRNP70-AID cell line constructed in this study were confirmed by PCR and Western blot.

Mycoplasma contamination

All cell lines have been tested for mycoplasma contamination free by PCR.

Commonly misidentified lines (See [ICLAC](#) register)

No cell lines used in this study were found in the database of commonly misidentified cell lines that is maintained by ICLAC and NCB BioSample.

ChIP-seq

Data deposition

- ☒ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☒ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links

May remain private before publication.

<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE134287> (token: qvmvsqgoxjglxwj)

Files in database submission

70AID-0h_Pol2S2P.bed; 70AID-4h_Pol2S2P.bed; 70AID-0h_Pol2S5P.bed; 70AID-4h_Pol2S5P.bed

Genome browser session

(e.g. [UCSC](#))

Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.

Methodology

Replicates

ChIP-seq for each condition replicated once.

Sequencing depth

70AID-0h_Pol2S2P: total reads--8762112, uniq mapping ratio--75.41%, reads length--143nt, paired-end;
70AID-4h_Pol2S2P: total reads--8373124, uniq mapping ratio--73.72%, reads length--143nt, paired-end;
70AID-0h_Pol2S5P: total reads--7259811, uniq mapping ratio--90.16%, reads length--150nt, paired-end;
70AID-4h_Pol2S5P: total reads--7373872, uniq mapping ratio--90.49%, reads length--150nt, paired-end.

Antibodies

Ser2P Pol II: Cell Signaling Technology, 13499; Ser5P Pol II: Cell Signaling Technology, 13523

Peak calling parameters

bowtie2 -x ./mm10 -1 sample_R1.fastq -2 sample_R2.fastq -k 1 -S sample.sam
macs14 -t sample.sam -c input.sam -g mm -n sample_input -p 1e-5 -B -S

Data quality

70AID-0h_Pol2S2P: FC > 5, FDR < 5%, 5890 peaks;
70AID-4h_Pol2S2P: FC > 5, FDR < 5%, 7055 peaks;
70AID-0h_Pol2S5P: FC > 5, FDR < 5%, 14780 peaks;
70AID-4h_Pol2S5P: FC > 5, FDR < 5%, 16170 peaks.

Software

Bowtie2, Bedtools, macs, ngs.plot

Flow Cytometry

Plots

Confirm that:

- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

Cells expressing a GFP reporter fused with different insert were harvested, and perform FACS analysis directly to measure the GFP intensity change. The flow cytometry was used for a mini-screen, no plot shown. The summary of GFP intensity changes was shown in Extended Data Figure 3j.

Instrument

BDCalibur

Software

Data collection: BD FACSDiva 8.0; Data analysis: FlowJo 7.6.1

Cell population abundance

No sorting was conducted.

Gating strategy

FSC/SSC were used to discern single cells from doublets/multiple cells. Samples without GFP reporter transfection were used to establish boundaries between negative and positive cells.

- ☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.